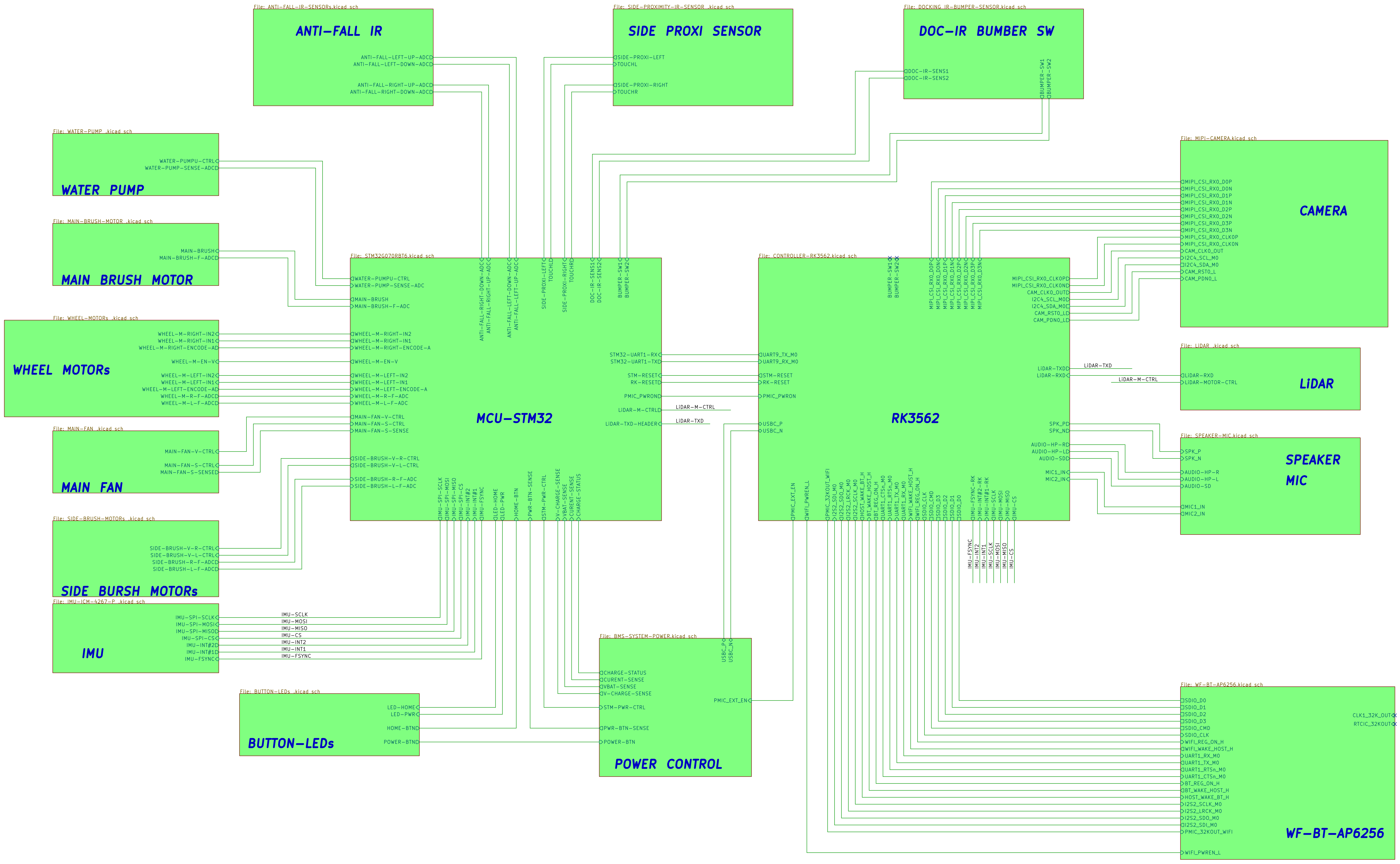
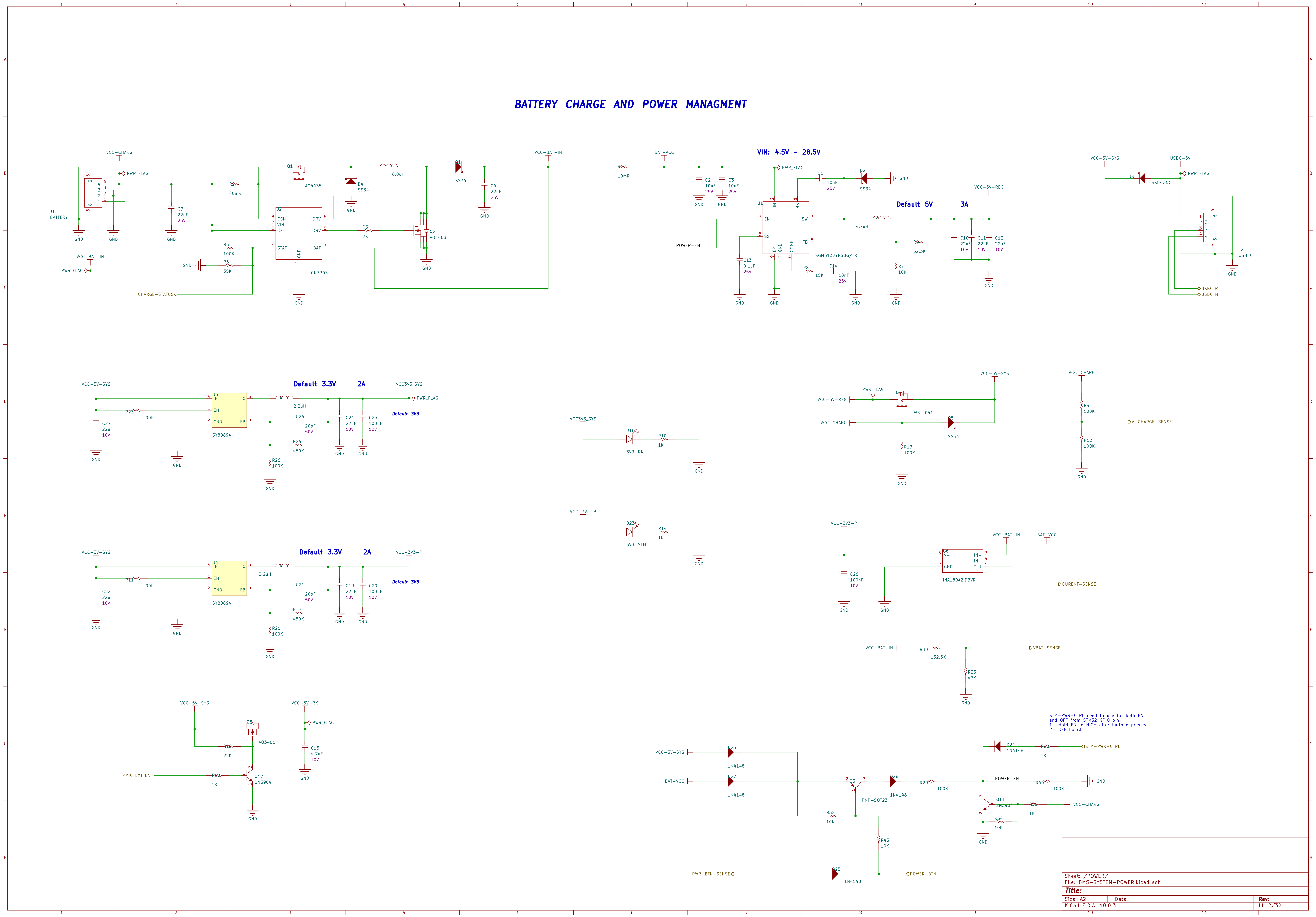
Controller-RK3562 CONTROLLER TOP

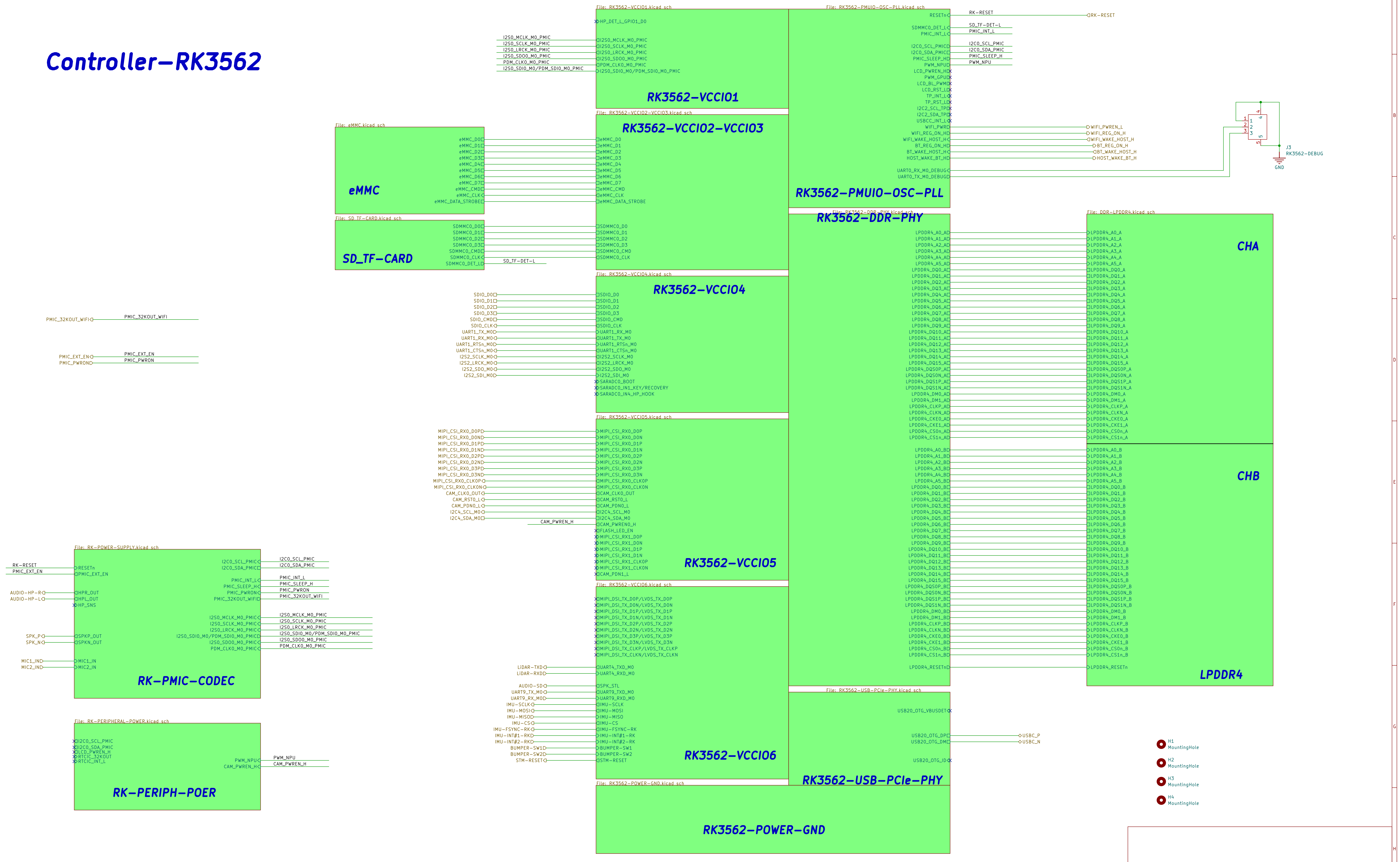


Sheet: /
File: oomwoo-kicad.kicad_sch

Title:

Size: A2 Date: Rev:
KiCad E.D.A. 10.0.3 Id: 1/32


BATTERY CHARGE AND POWER MANAGMENT
VIN: 4.5V ~ 28.5V
Default 5V 3A
Default 3.3V 2A
Default 3V3
STM-PWR-CTRL need to use for both EN
and OFF from STM32 GPIO pin.
1- Hold EN to HIGH after buttone pressed
2- OFF board
CN3303
SGM6132YPS8G/TR
SY8089A
AO4435
AO4468
AO3401
WST4041
INA180A2IDBVR
PNP-SOT23
2N3904
1N4148
SS34
SS54
SS54/NC
USB C
BATTERY
VCC-CHARG
VCC-BAT-IN
BAT-VCC
VCC-5V-SYS
VCC-5V-REG
VCC3V3_SYS
VCC-3V3-P
VCC-5V-RK
USBC-5V
PWR_FLAG
POWER-EN
CHARGE-STATUS
V-CHARGE-SENSE
CURENT-SENSE
VBAT-SENSE
PWR-BTN-SENSE
POWER-BTN
STM-PWR-CTRL
PMIC_EXT_END
USBC_P
USBC_N
3V3-RK
3V3-STM
GND
Sheet: /POWER/
File: BMS-SYSTEM-POWER.kicad_sch
Title:
Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.3
Id: 2/32



Sheet: /RK3562/
File: CONTROLLER-RK3562.kicad_sch

Title:

Size: A2 | Date: | Rev:

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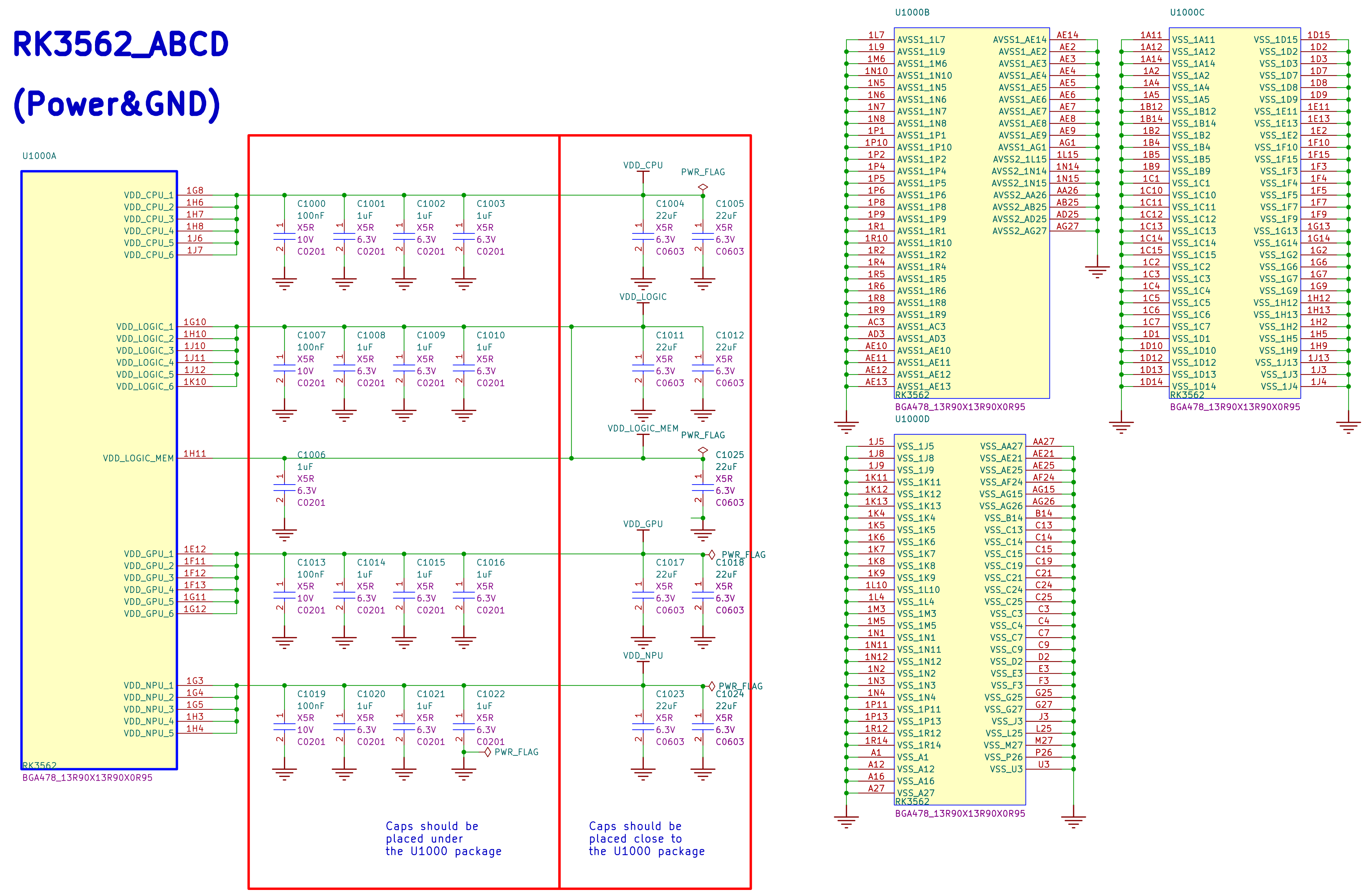
RK3562_ABCD

(Power&GND)

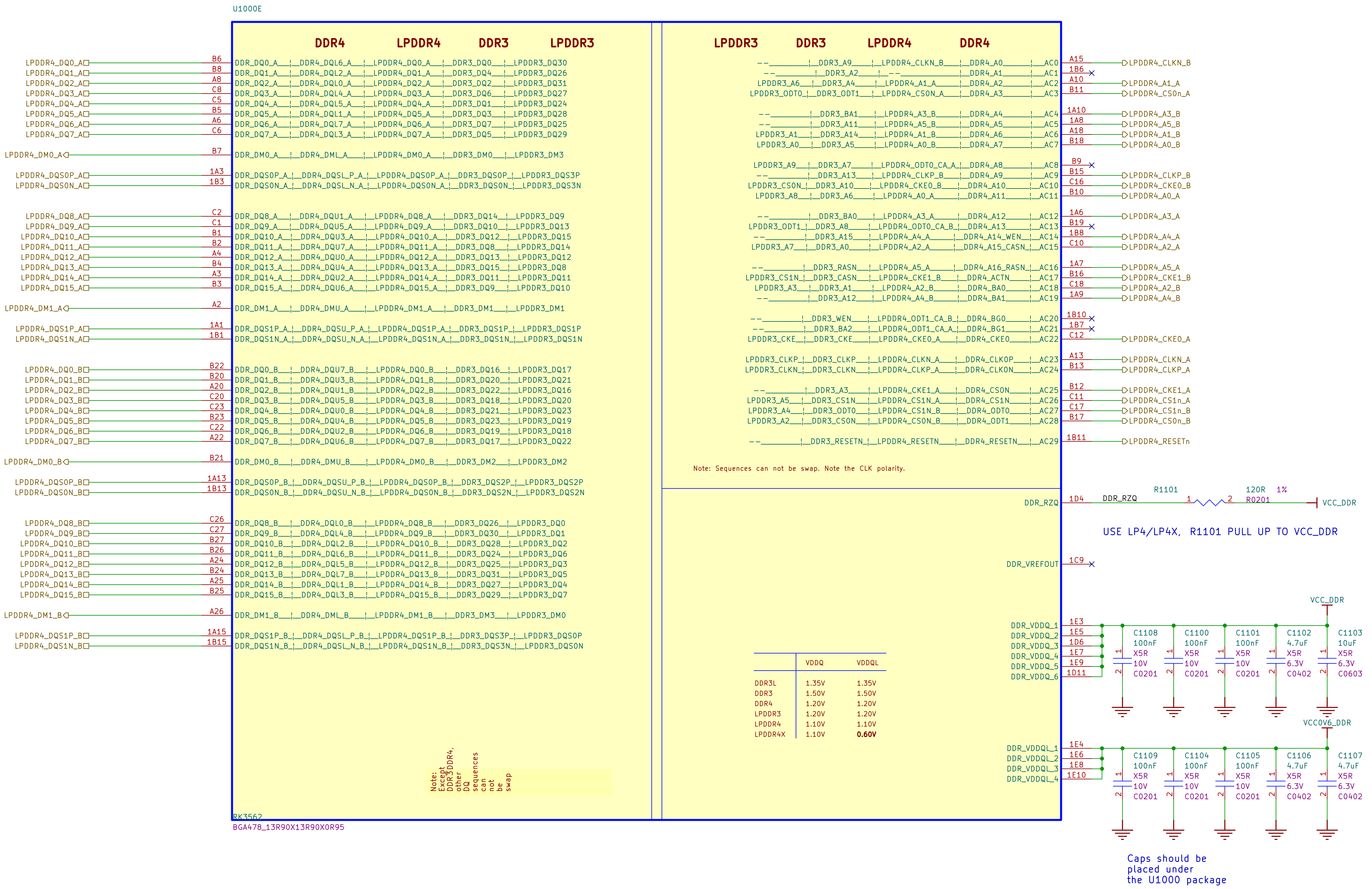


Sheet: /RK3562/RK3562-POWER/
File: RK3562-POWER-GND.kicad_sch

Title:

Size: A2 Date: Rev:
KiCad E.D.A. 10.0.3 Id: 4/32

RK3562_E(DDRPHY)



	VDDQ	VDDQL
DDR3L	1.35V	1.35V
DDR3	1.50V	1.50V
DDR4	1.20V	1.20V
LPDDR3	1.20V	1.20V
LPDDR4	1.10V	1.10V
LPDDR4X	1.10V	**0.60V**

Sheet: /RK3562/RK3562-DDR-PHY/
File: RK3562-DDR-PHY.kicad_sch

Title:

Size: A2 | Date: | **Rev:**

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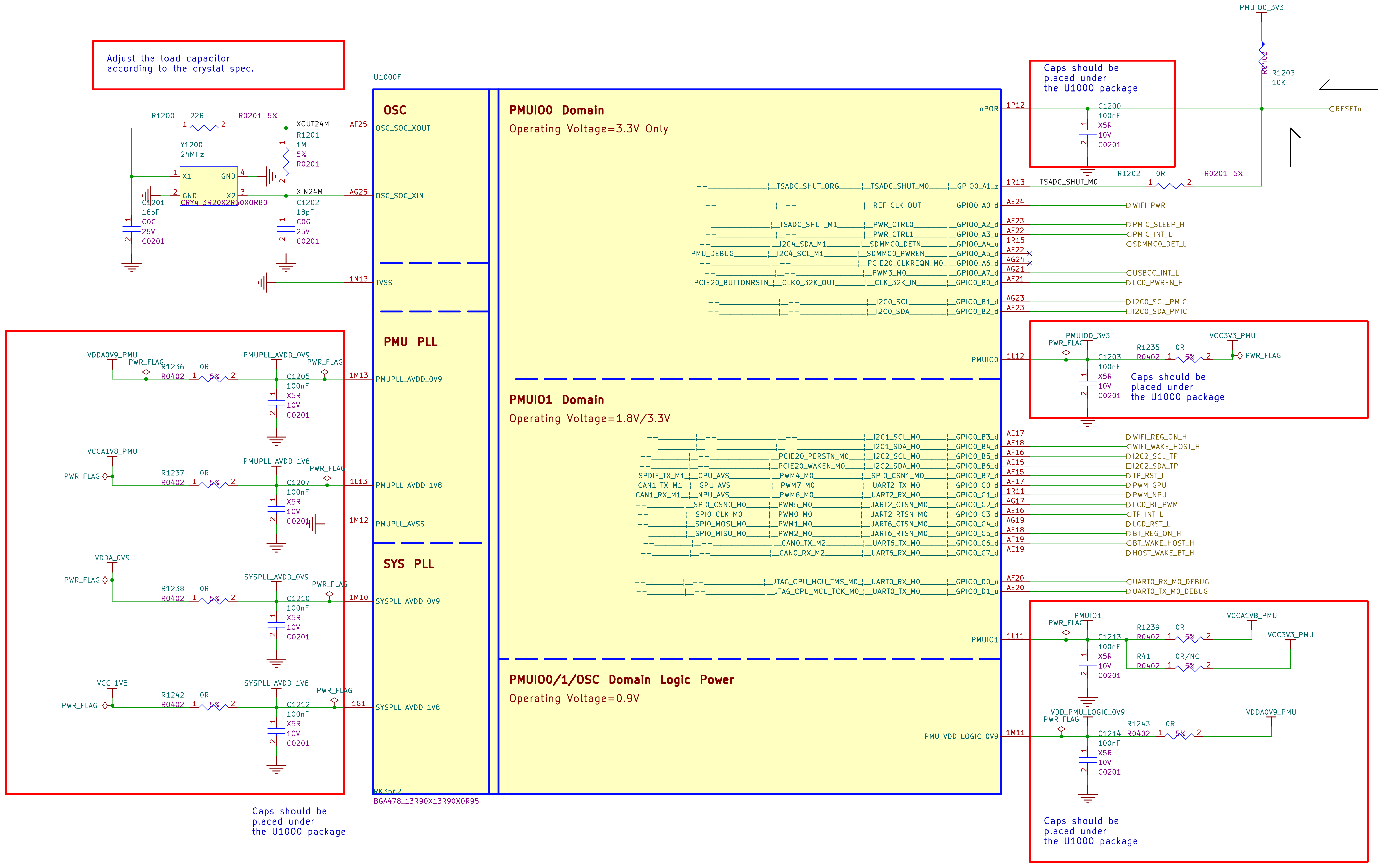
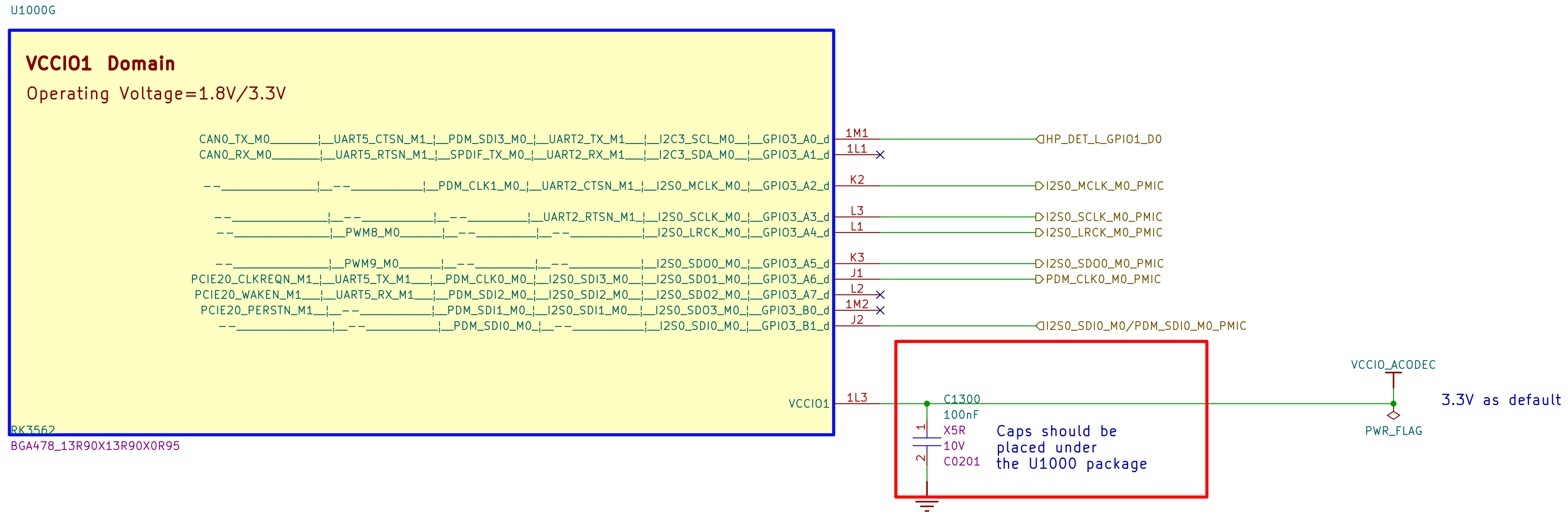
RK3562_F(PMUIO/OSC/PLL)



RK3562_G(VCCIO1 Domain)



Sheet: /RK3562/RK3562-VCCIO1/
File: RK3562-VCCIO1.kicad_sch

Title:

Size: A2 | Date: | **Rev:**

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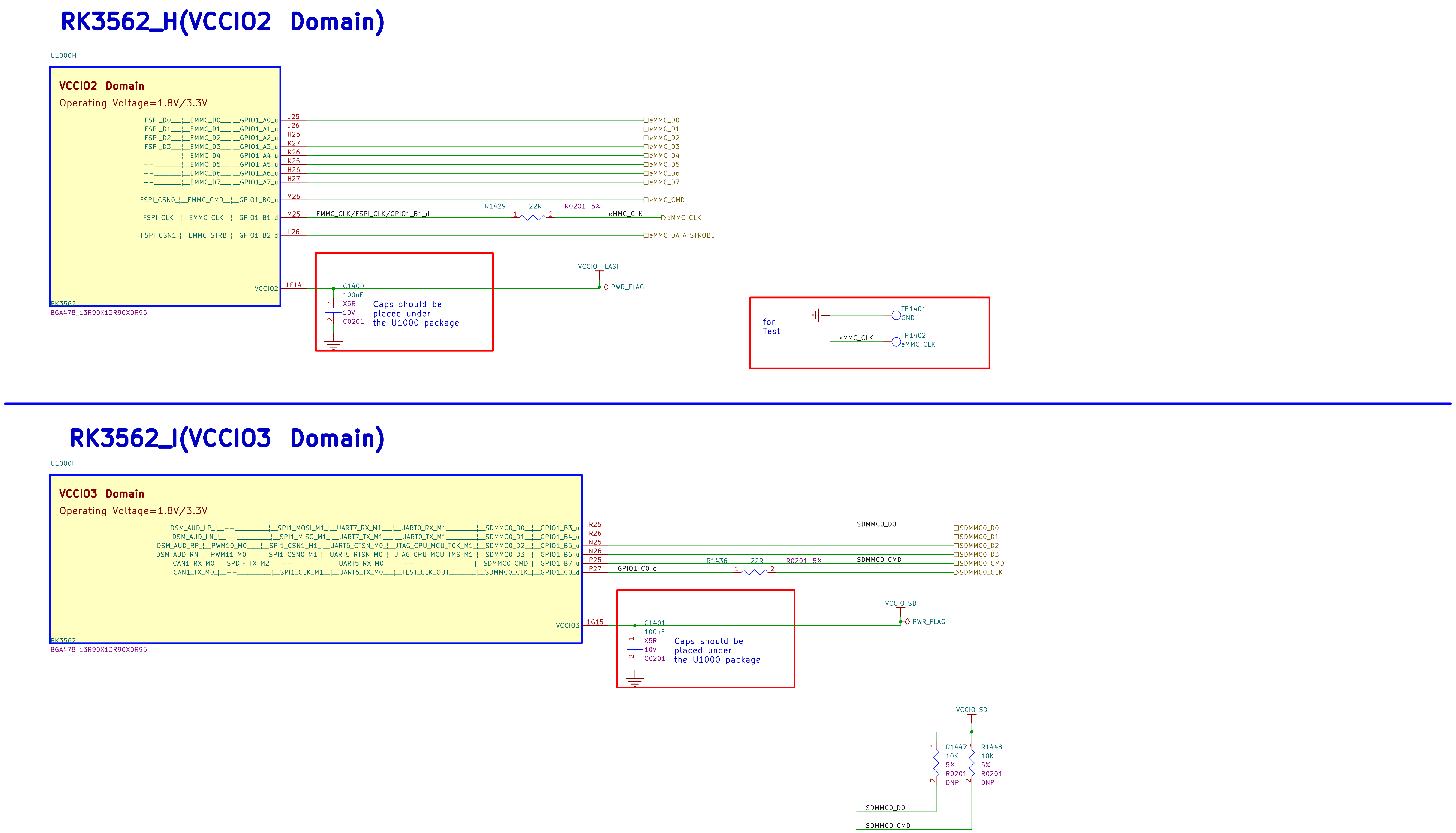

Sheet: /RK3562/RK3562-VCCIO2-VCCIO3/
File: RK3562-VCCIO2-VCCIO3.kicad_sch

Title:

Size: A2 | Date: | Rev:

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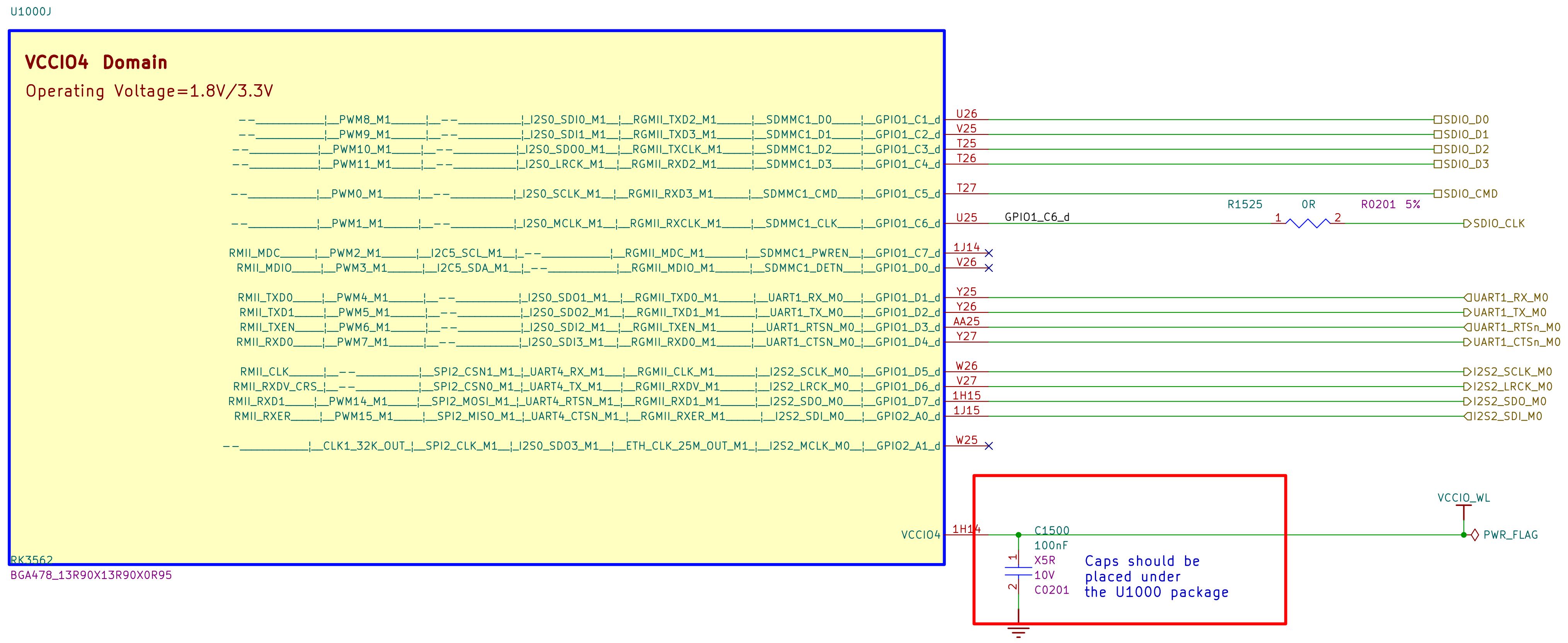
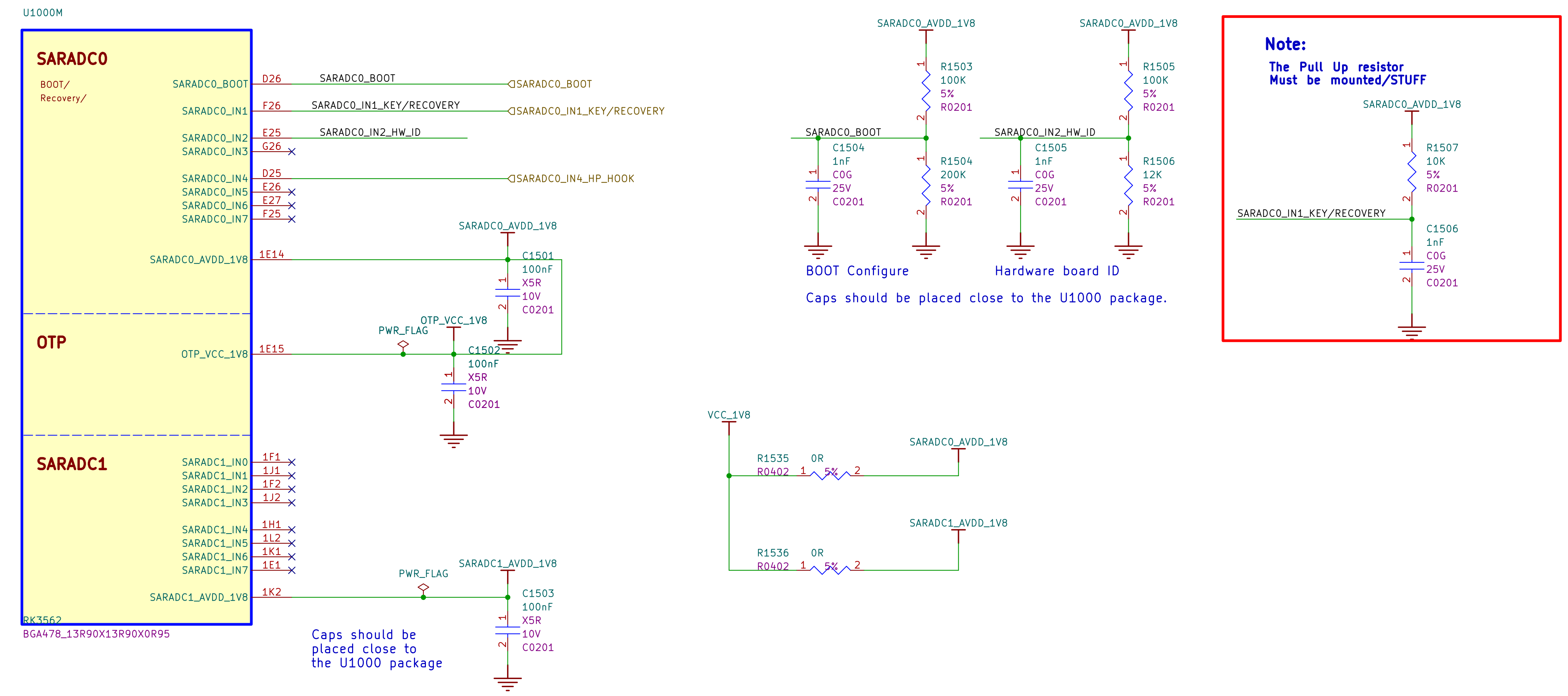
RK3562_J(VCCIO4 Domain)

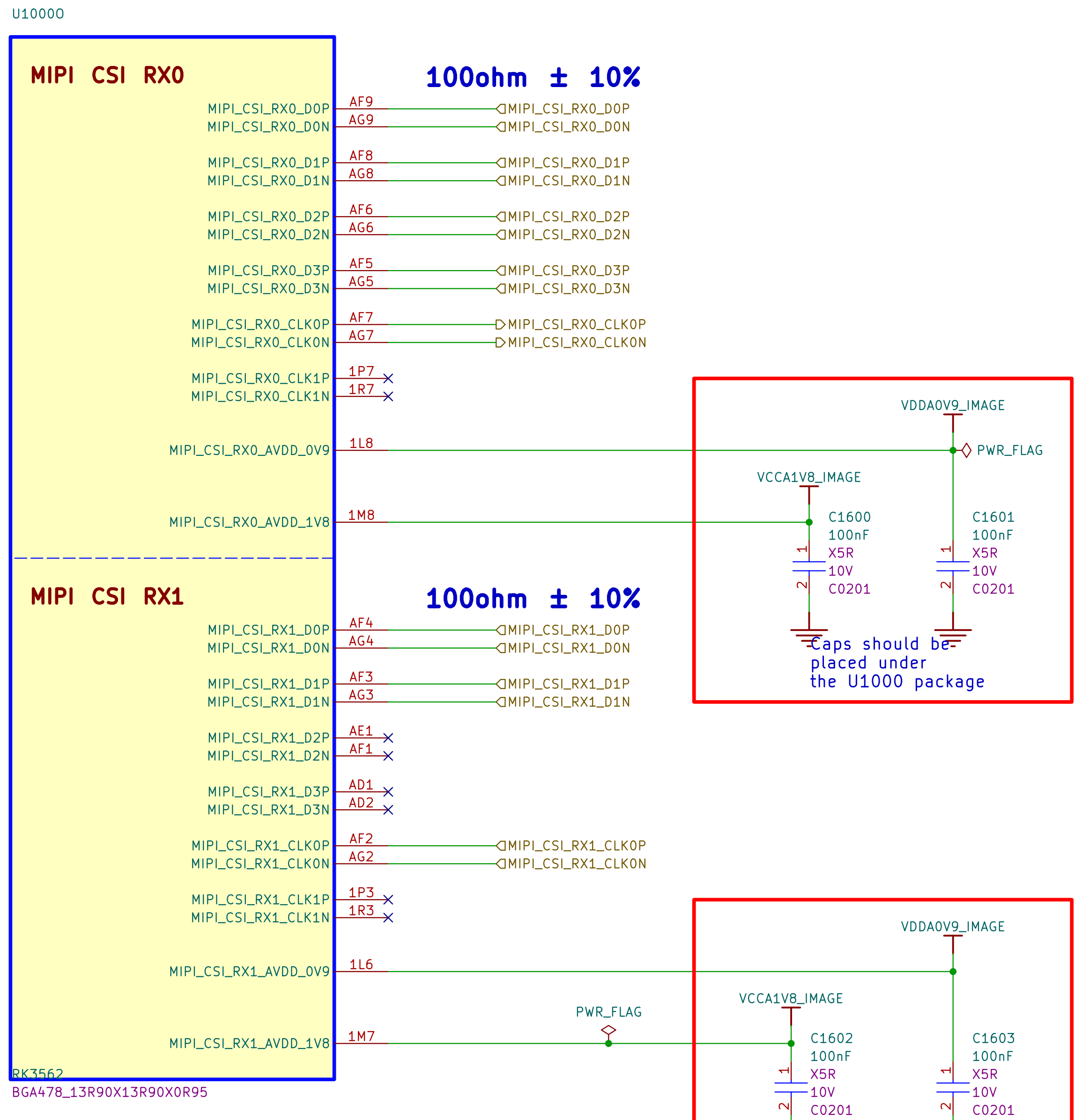


Config Table for SARADC0_BOOT				
Item	**Rup**	**Rdown**	**ADC Value**	**Boot Mode**
Config1	NC	100K	0	USB (Maskrom mode)
Config2	100K	12K	114	
Config3	100K	27K	228	FSPI--USB
Config4	100K	51K	342	
Config5	100K	82K	456	
Config6	100K	120K	570	EMMC--USB
Config7	100K	200K	683	EMMC--SD Card--USB
Config8	100K	330K	796	SD Card--USB
Config9	100K	820K	910	
Config10	100K	NC	1023	FSPI--EMMC--SD Card--USB

RK3562_M(SARADC/OTP)



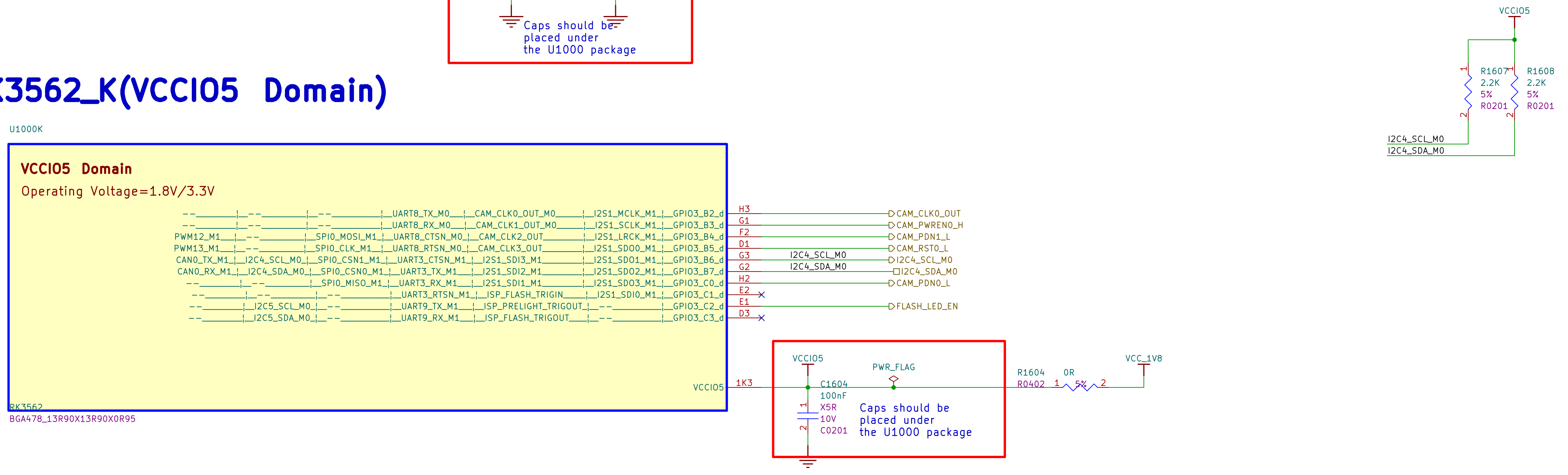
RK3562_O(MIPI_CSI_RX0 & RX1)



Usage of each MIPI CSI RXn & CLKn

Option1	Sensor1	x4Lane	MIPI_CSI_RXn_D0-3 MIPI_CSI_RXn_CLK0
Option2	Sensor1	x2Lane	MIPI_CSI_RXn_D0-1 MIPI_CSI_RXn_CLK0
	+ Sensor2	x2Lane	MIPI_CSI_RXn_D2-3 MIPI_CSI_RXn_CLK1

RK3562_K(VCCIO5 Domain)

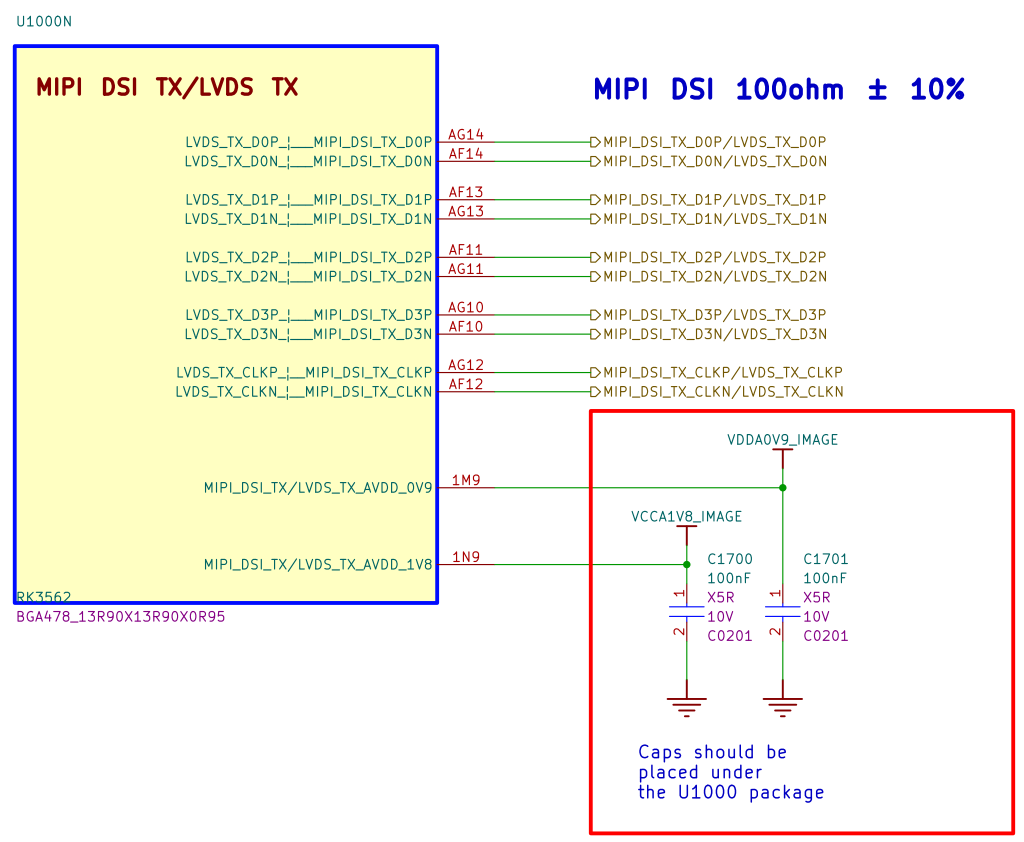


Sheet: /RK3562/RK3562-VCCIO5/
File: RK3562-VCCIO5.kicad_sch

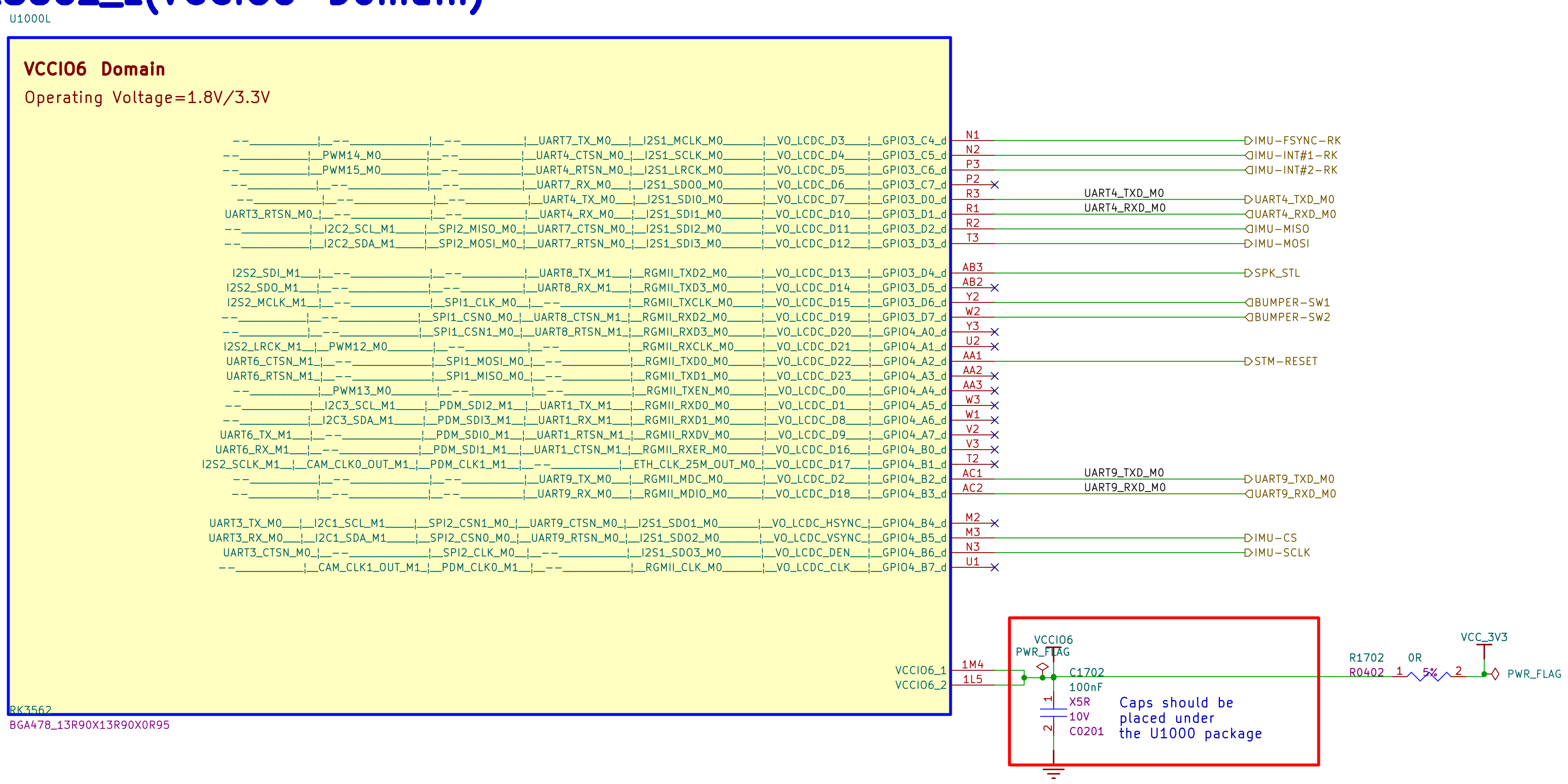
Title:

Size: A2 Date: Rev:
KiCad E.D.A. 10.0.3 Id: 10/32

RK3562_N(MIPI_DSI_TX/LVDS_TX)



RK3562_L(VCCIO6 Domain)

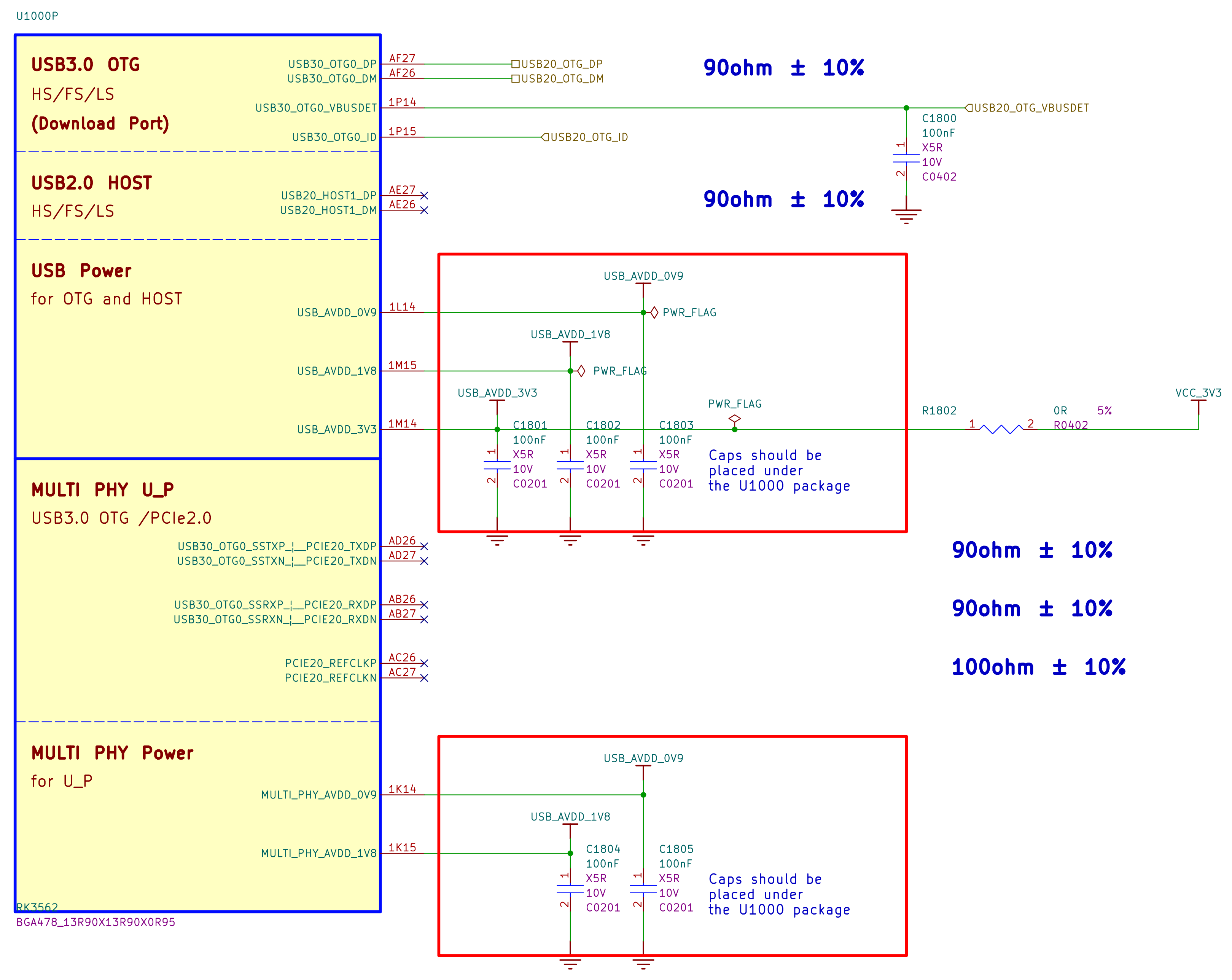


Sheet: /RK3562/RK3562-VCCIO6/
File: RK3562-VCCIO6.kicad_sch

Title:

Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 11/32

RK3562_P(USB/PCIe)

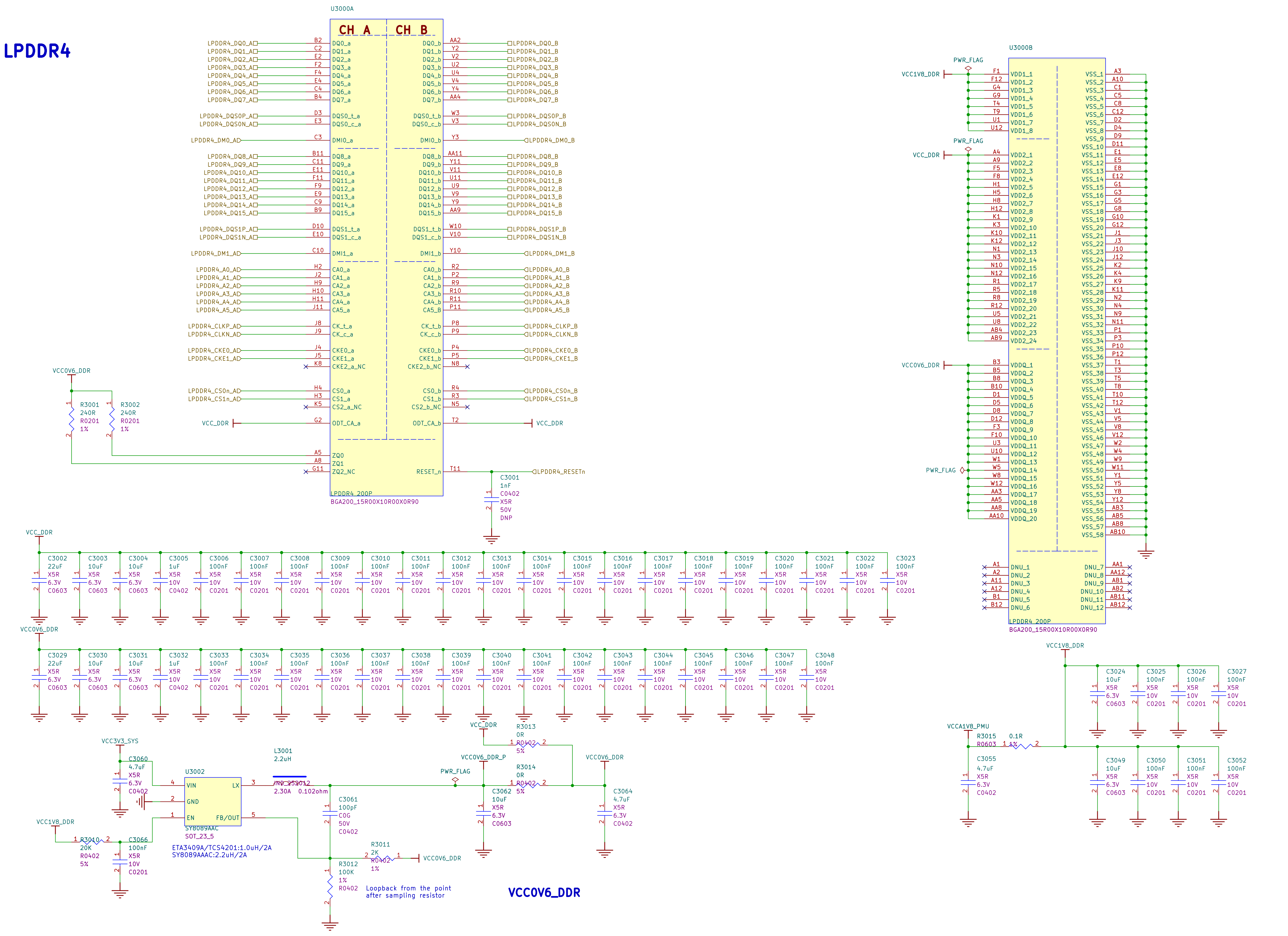


Sheet: /RK3562/RK3562-USB-PCIe-PHY/
File: RK3562-USB-PCIe-PHY.kicad_sch

Title:

Size: A2 | Date: | **Rev:**

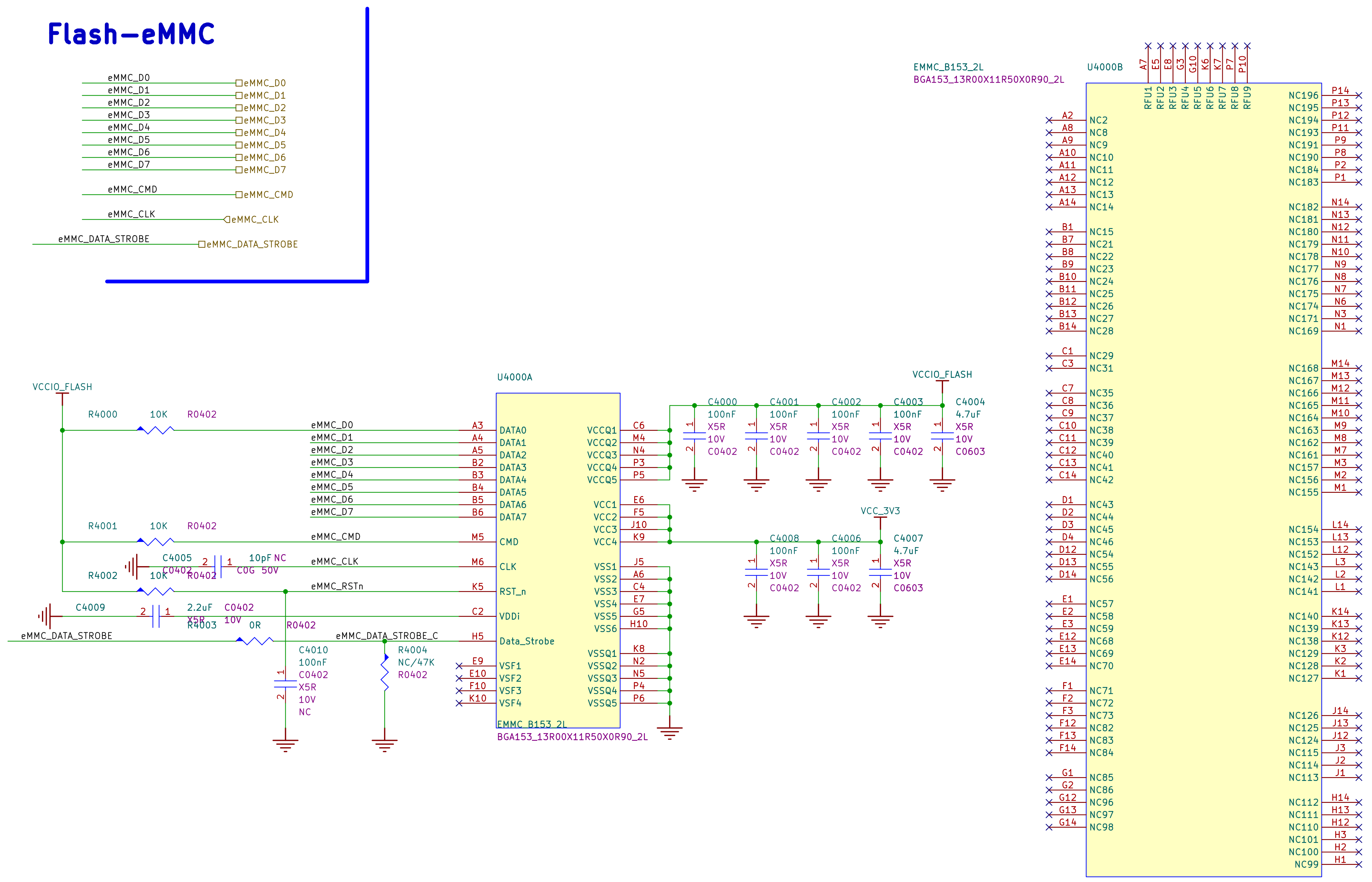
KiCad E.D.A. 10.0.3 | Id: 12/32



Sheet: /RK3562/DRAM-LPDDR4/
File: DDR-LPDDR4.kicad_sch

Title:

Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.3		Id: 13/32



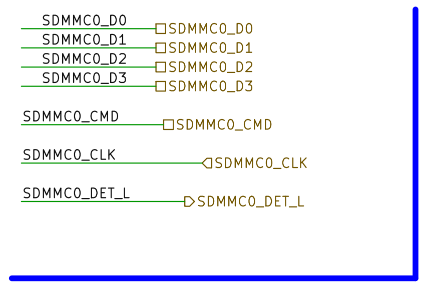
Sheet: /RK3562/eMMC/
File: eMMC.kicad_sch

Title:

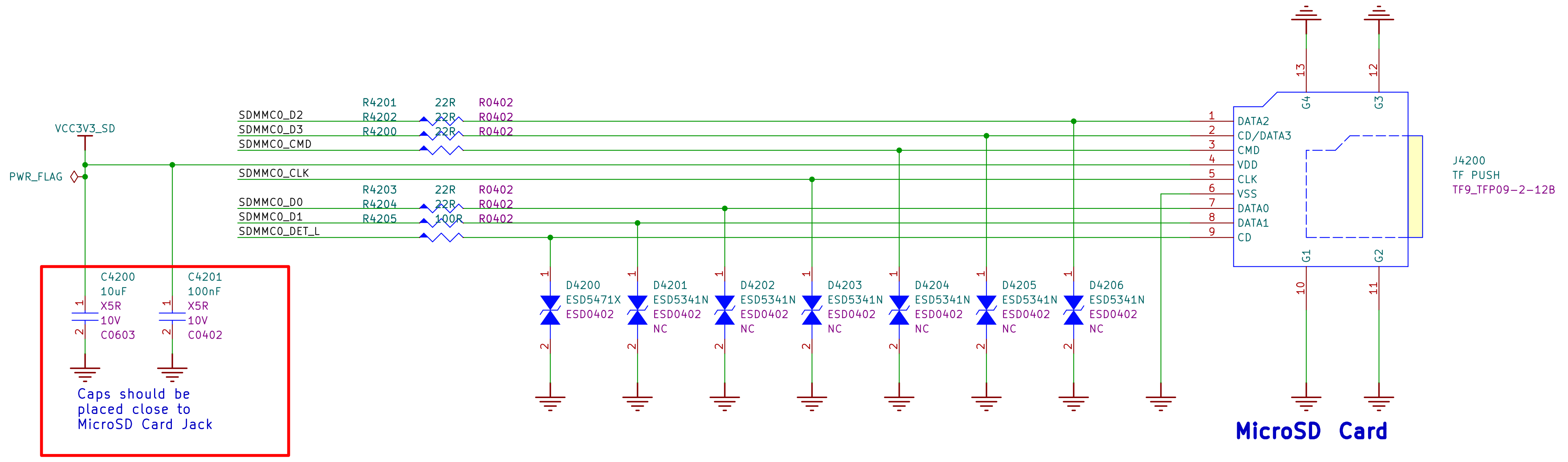
Size: A2 Date: Rev:

KiCad E.D.A. 10.0.3 Id: 14/32

Flash-MicroSD_Card



Note:
Take care of the type of MicroSD Card Jack,
The type here follow the logic below:
DET is float @ card ejected; the internal pull up decided the IO pin status
DET is connected to GND @ card inserted
If other type of Jack is used, need to shift the status of DET.

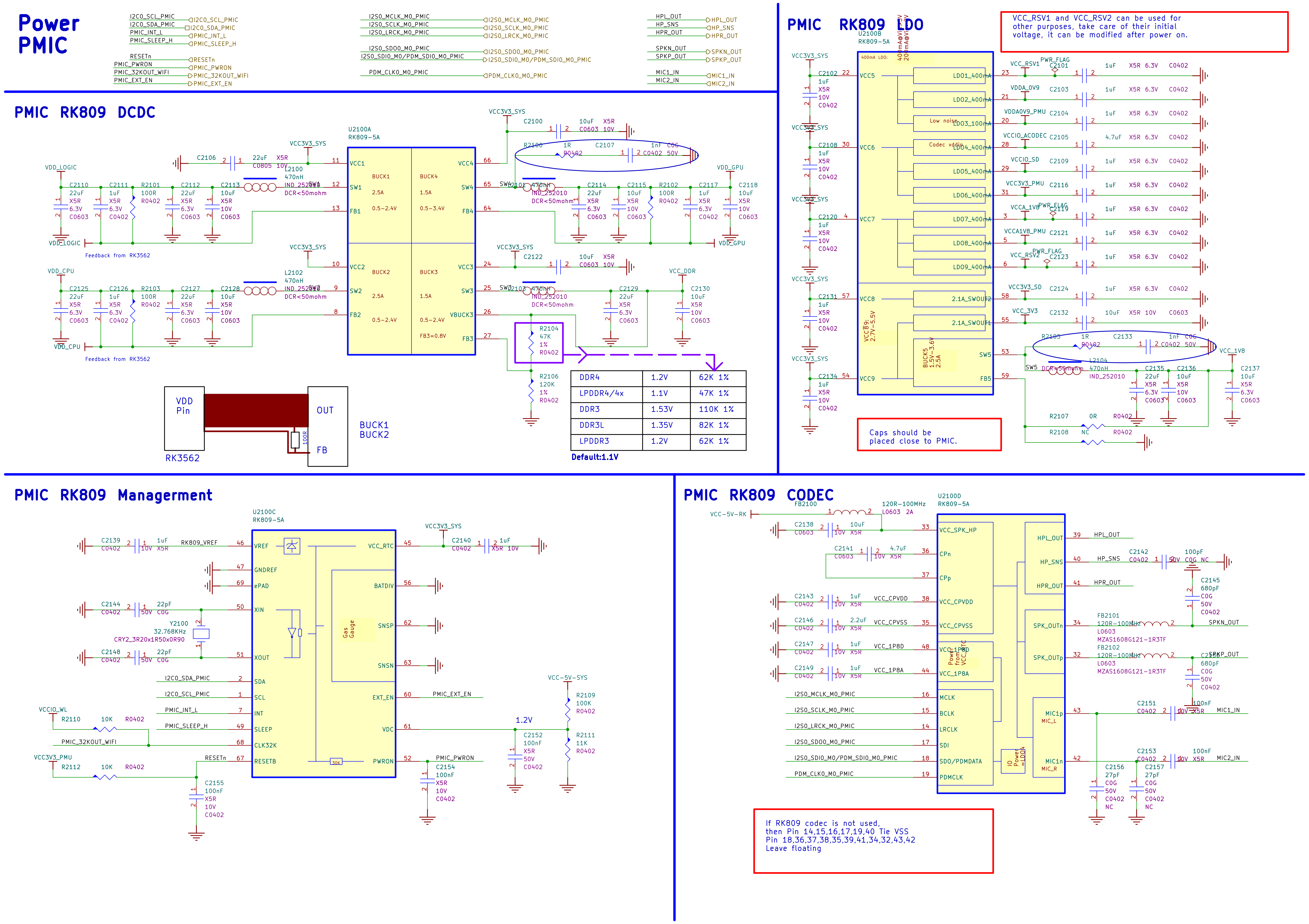


Sheet: /RK3562/SD_TF CARD/
File: SD_TF-CARD.kicad_sch

Title:

Size: A2 | Date: | Rev:

KiCad E.D.A. 10.0.3 | Id: 15/32



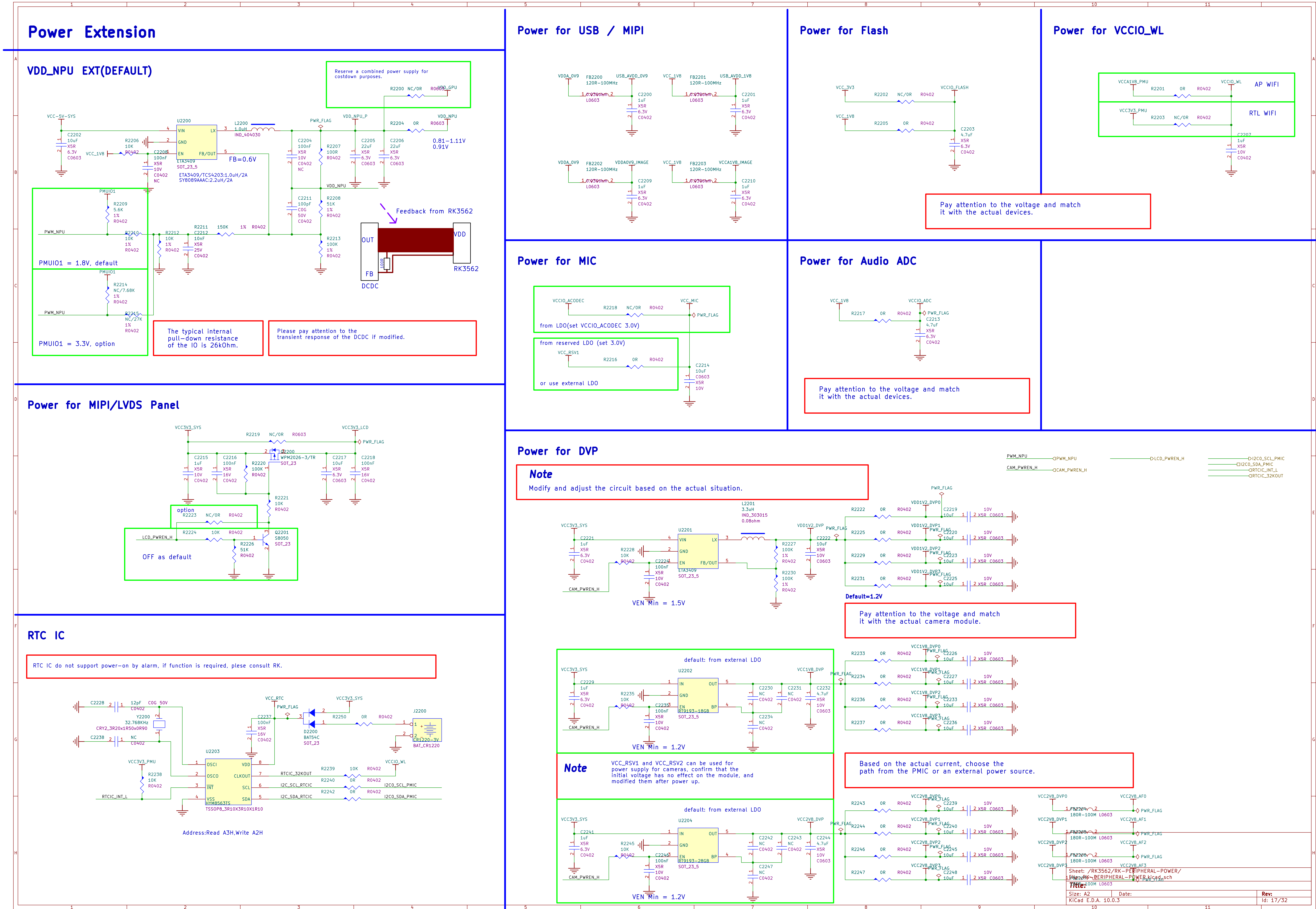
DDR4	1.2V	62K 1%
LPDDR4/4x	1.1V	47K 1%
DDR3	1.53V	110K 1%
DDR3L	1.35V	82K 1%
LPDDR3	1.2V	62K 1%

Default:1.1V

Sheet: /RK3562/RK-POWER-SUPPLY/
File: RK-POWER-SUPPLY.kicad_sch

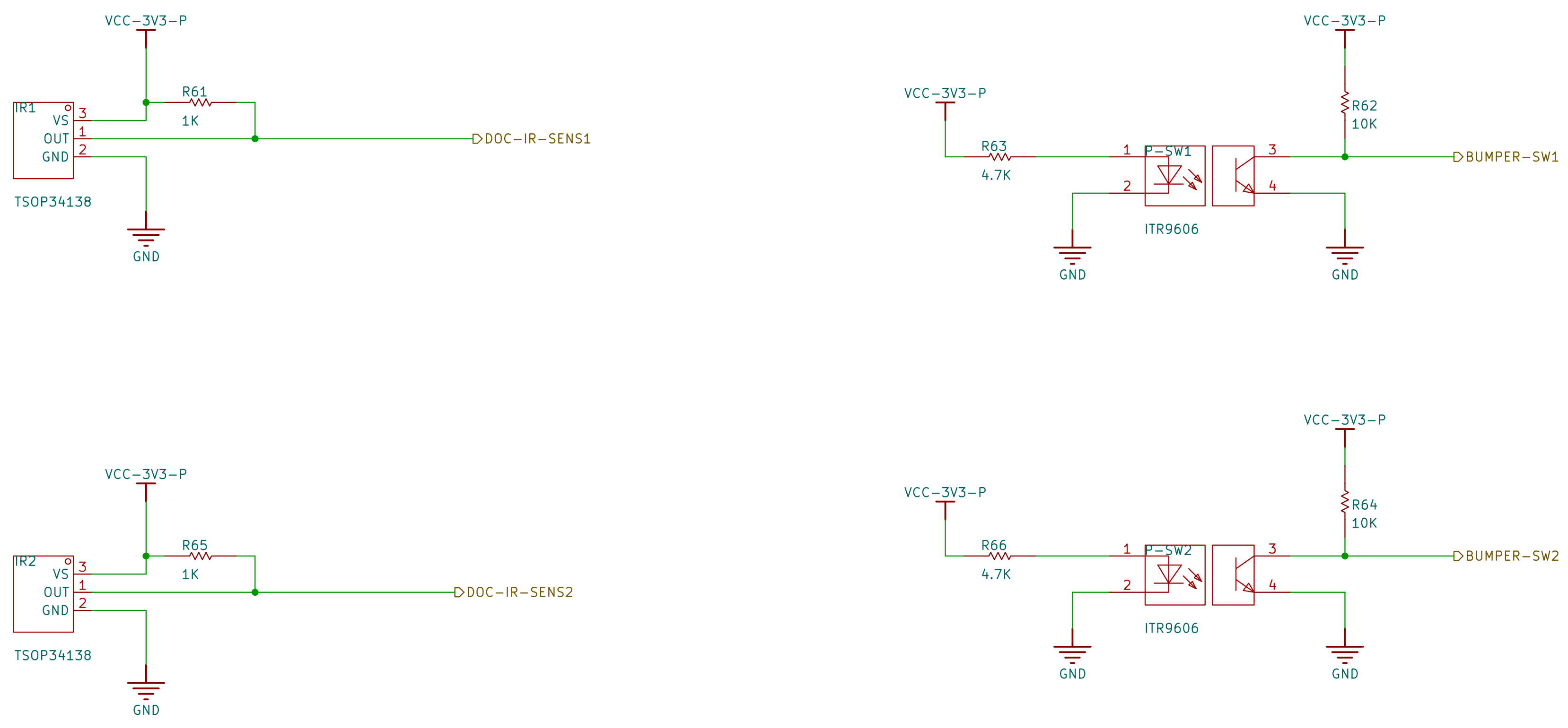
Title:

Size: A2 Date: Rev:
KiCad E.D.A. 10.0.3 Id: 16/32



Power Extension
VDD_NPU EXT(DEFAULT)
Reserve a combined power supply for costdown purposes.
FB=0.6V
0.81~1.11V
0.91V
ETA3409/TCS4203:1.0uH/2A
SY8089AAAC:2.2uH/2A
Feedback from RK3562
OUT
FB
DCDC
VDD
RK3562
PMUIO1 = 1.8V, default
PMUIO1 = 3.3V, option
The typical internal pull-down resistance of the IO is 26kOhm.
Please pay attention to the transient response of the DCDC if modified.
Power for USB / MIPI
Power for Flash
Power for VCCIO_WL
AP WIFI
RTL WIFI
Pay attention to the voltage and match it with the actual devices.
Power for MIC
from LDO(set VCCIO_ACODEC 3.0V)
from reserved LDO (set 3.0V)
or use external LDO
Power for Audio ADC
Pay attention to the voltage and match it with the actual devices.
Power for MIPI/LVDS Panel
option
OFF as default
Power for DVP
Note
Modify and adjust the circuit based on the actual situation.
VEN Min = 1.5V
Default=1.2V
Pay attention to the voltage and match it with the actual camera module.
default: from external LDO
VEN Min = 1.2V
Note
VCC_RSV1 and VCC_RSV2 can be used for power supply for cameras, confirm that the initial voltage has no effect on the module, and modified them after power up.
Based on the actual current, choose the path from the PMIC or an external power source.
default: from external LDO
VEN Min = 1.2V
RTC IC
RTC IC do not support power-on by alarm, if function is required, plese consult RK.
Address:Read A3H,Write A2H
Sheet: /RK3562/RK-PERIPHERAL-POWER/
File: RK-PERIPHERAL-POWER.kicad_sch
Title:
Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.3
Id: 17/32

DOCKING–IR–BUMBER–SENSORs

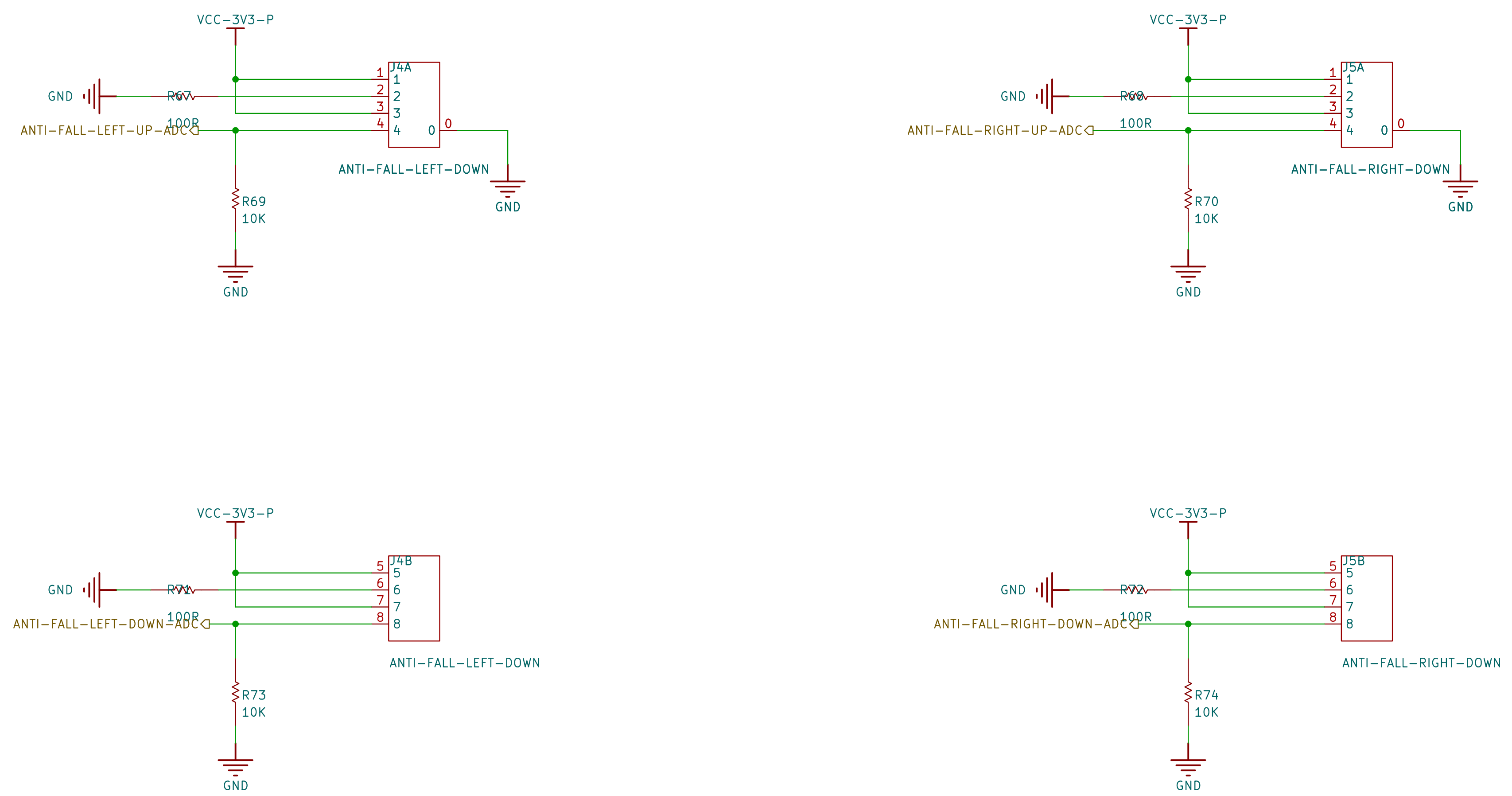


Sheet: /DOCKING–BUMBER/
File: DOCKING_IR–BUMPER–SENSOR.kicad_sch

Title:

Size: A2 Date: **Rev:**
KiCad E.D.A. 10.0.3 Id: 18/32

ANTI FALL SENSORs



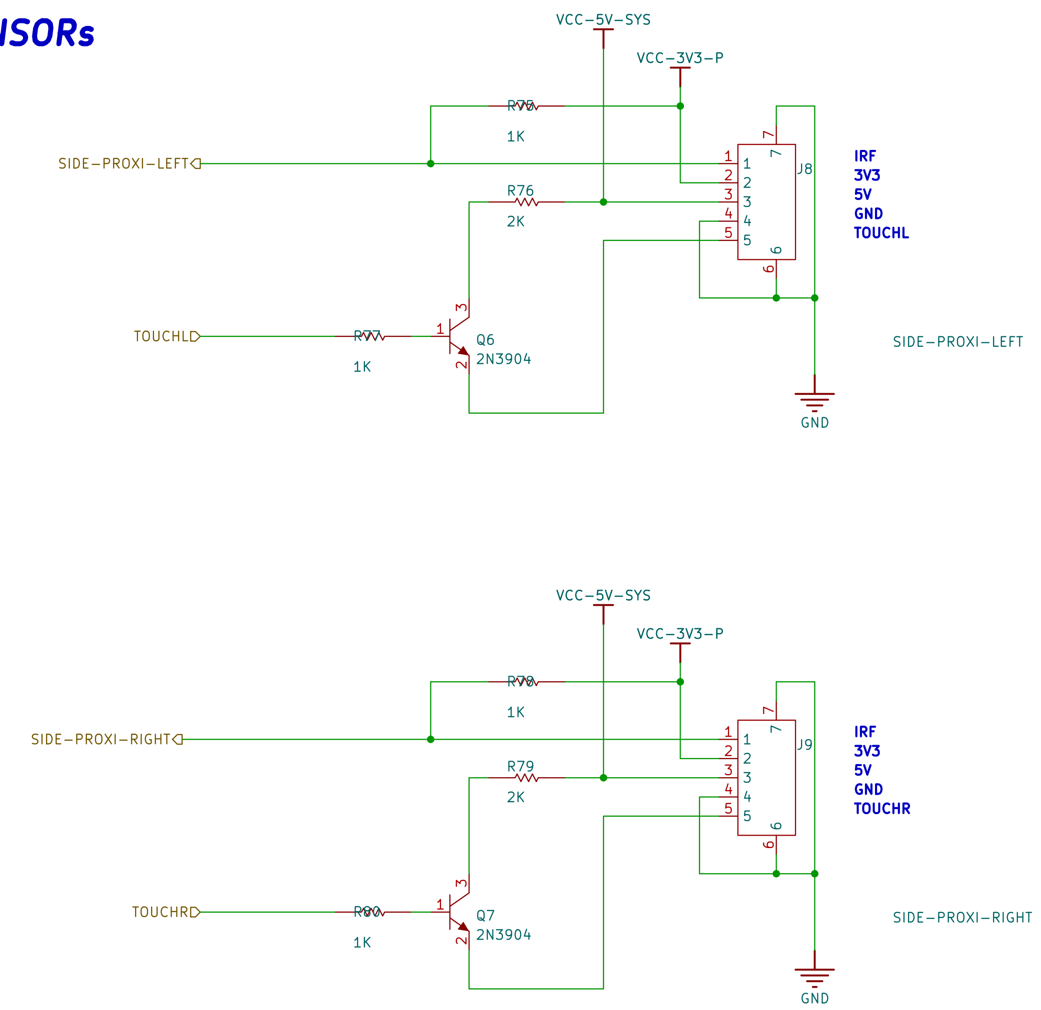
Sheet: /ANTI-FALL/
File: ANTI-FALL-IR-SENSORs.kicad_sch

Title:

Size: A2 | Date: | Rev:

KiCad E.D.A. 10.0.3 | Id: 19/32

SIDE PROXI IR SENSORs

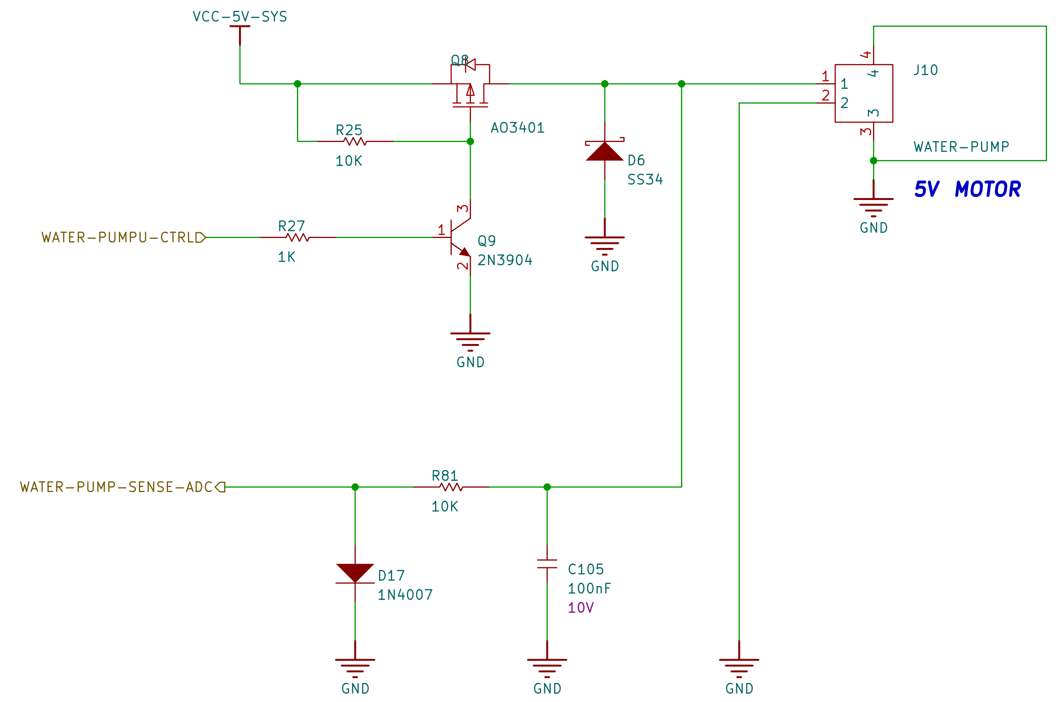


Sheet: /SIDE-PROXI/
File: SIDE-PROXIMITY-IR-SENSOR .kicad_sch

Title:

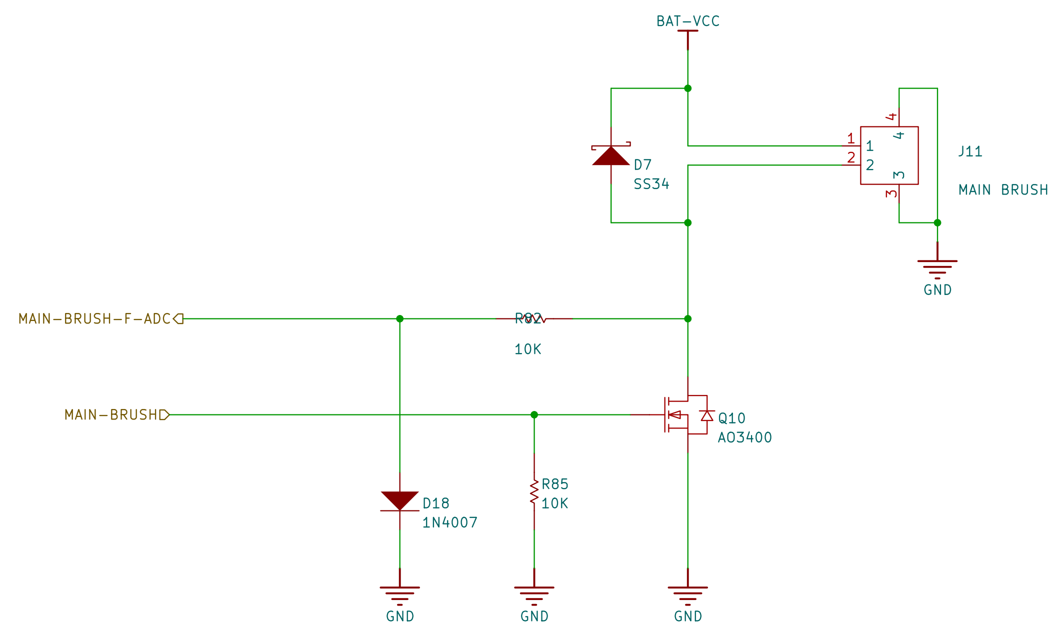
Size: A2 | Date: | Rev:
KiCad E.D.A. 10.0.3 | Id: 20/32

WATER PUMP MOTOR CONTROL



Sheet: /WATER-PUMP/
File: WATER-PUMP.kicad_sch
Title:
Size: A2 Date: **Rev:**
KiCad E.D.A. 10.0.3 Id: 21/32

MAIN BUSH MOTOR CONTROL

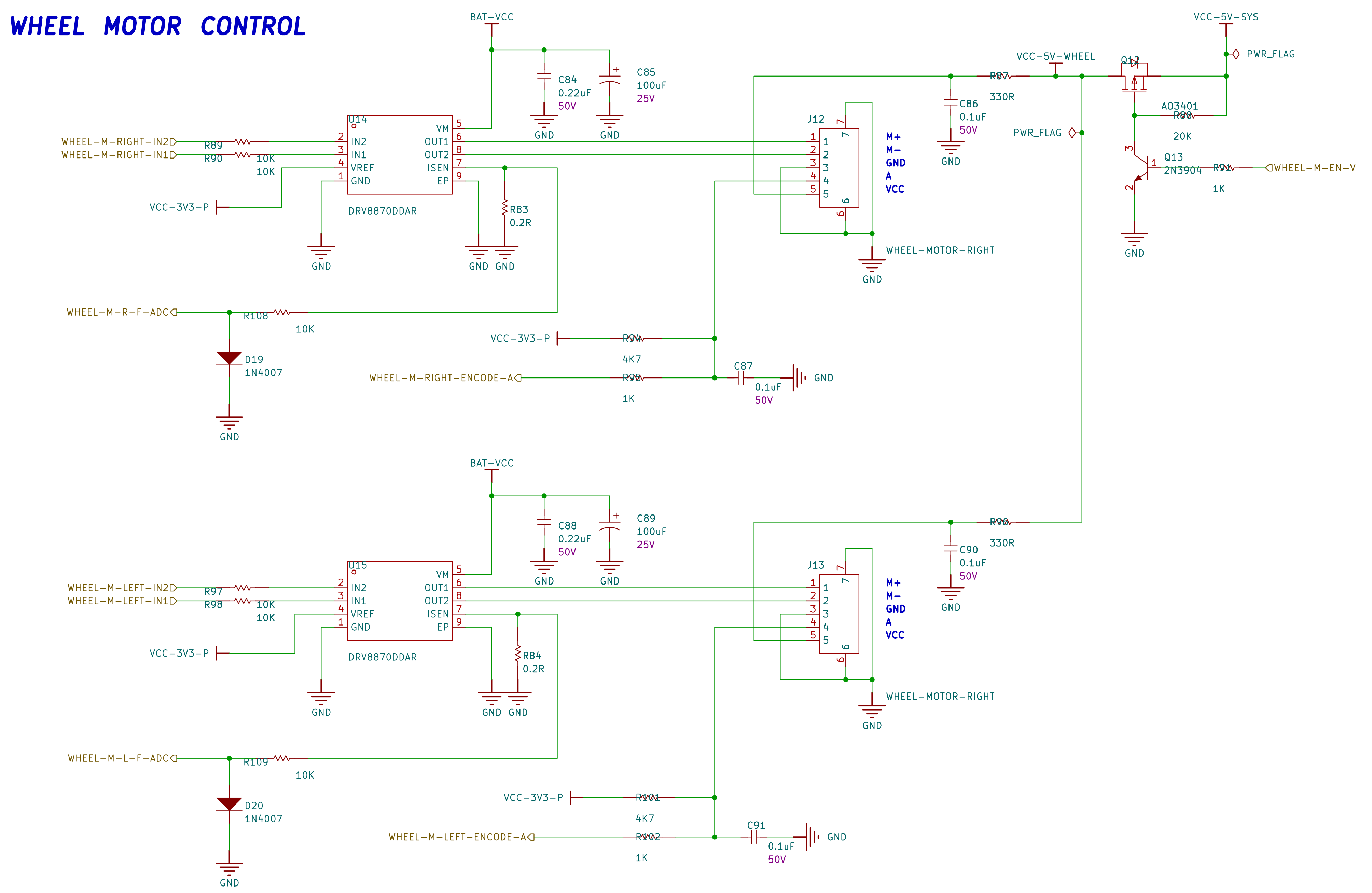


Sheet: /MAIN-BRUSH MOTOR/
File: MAIN-BRUSH-MOTOR .kicad_sch

Title:

Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.3		Id: 22/32

WHEEL MOTOR CONTROL



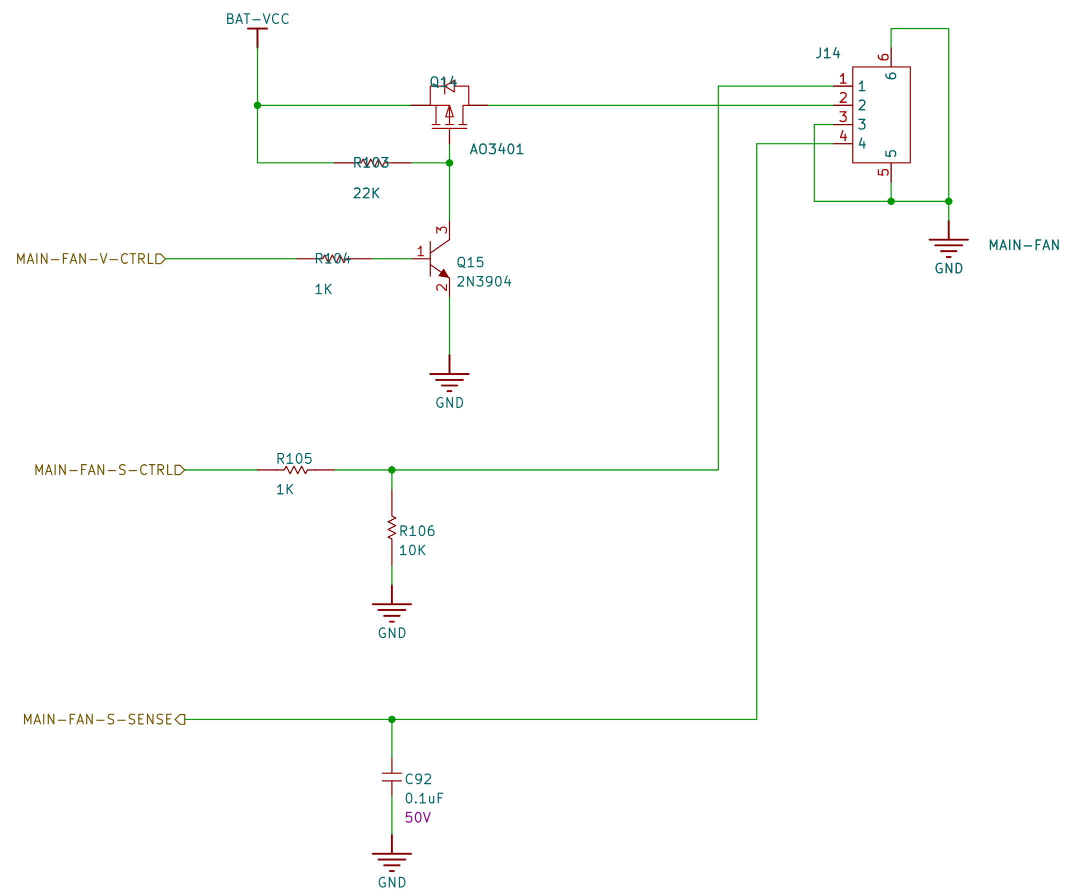
Sheet: /WHEEL MOTOR/
File: WHEEL-MOTORs .kicad_sch

Title:

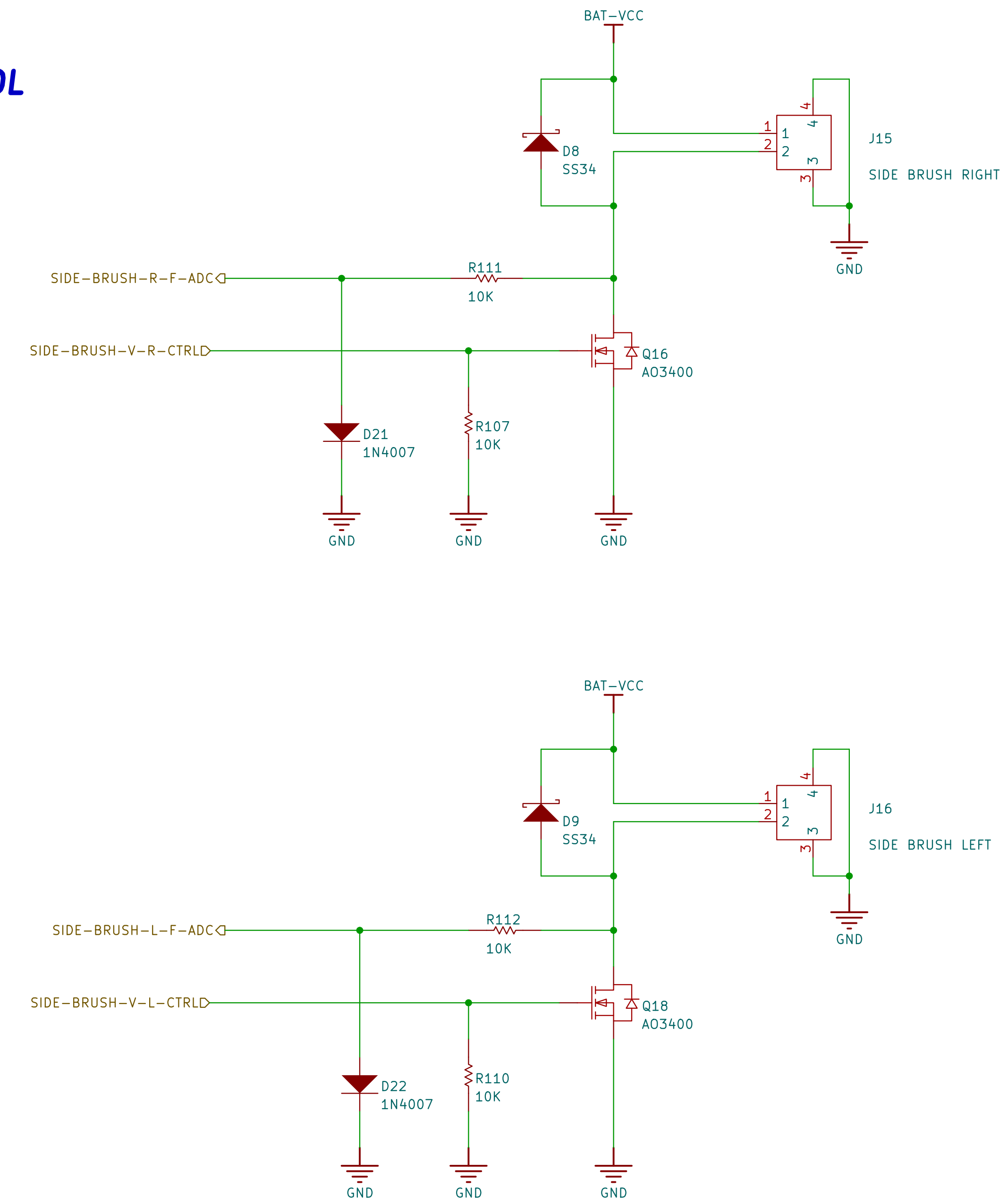
Size: A2 | Date: | Rev:

KiCad E.D.A. 10.0.3 | Id: 23/32

MAIN FAN CONTROL



SIDE BRUSH MOTOR CONTROL

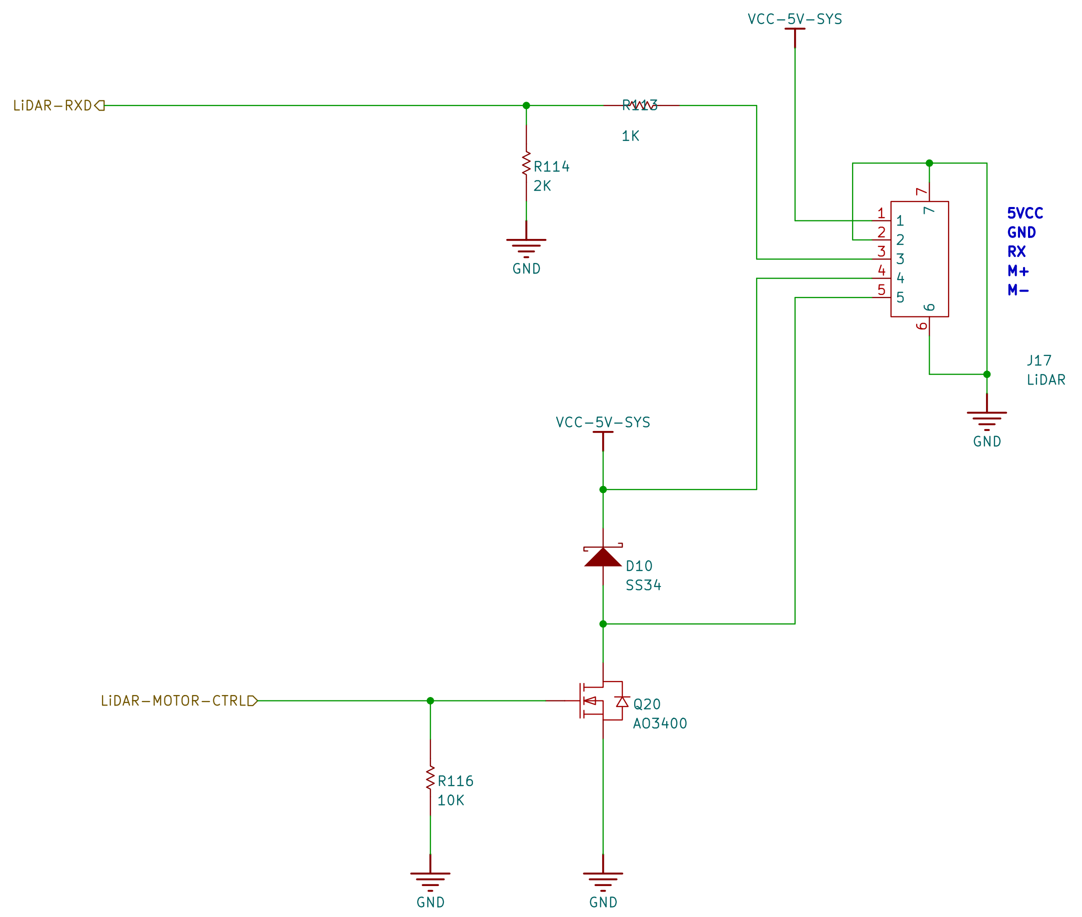


Sheet: /SIDE-BRUSH/
File: SIDE-BRUSH-MOTORs .kicad_sch

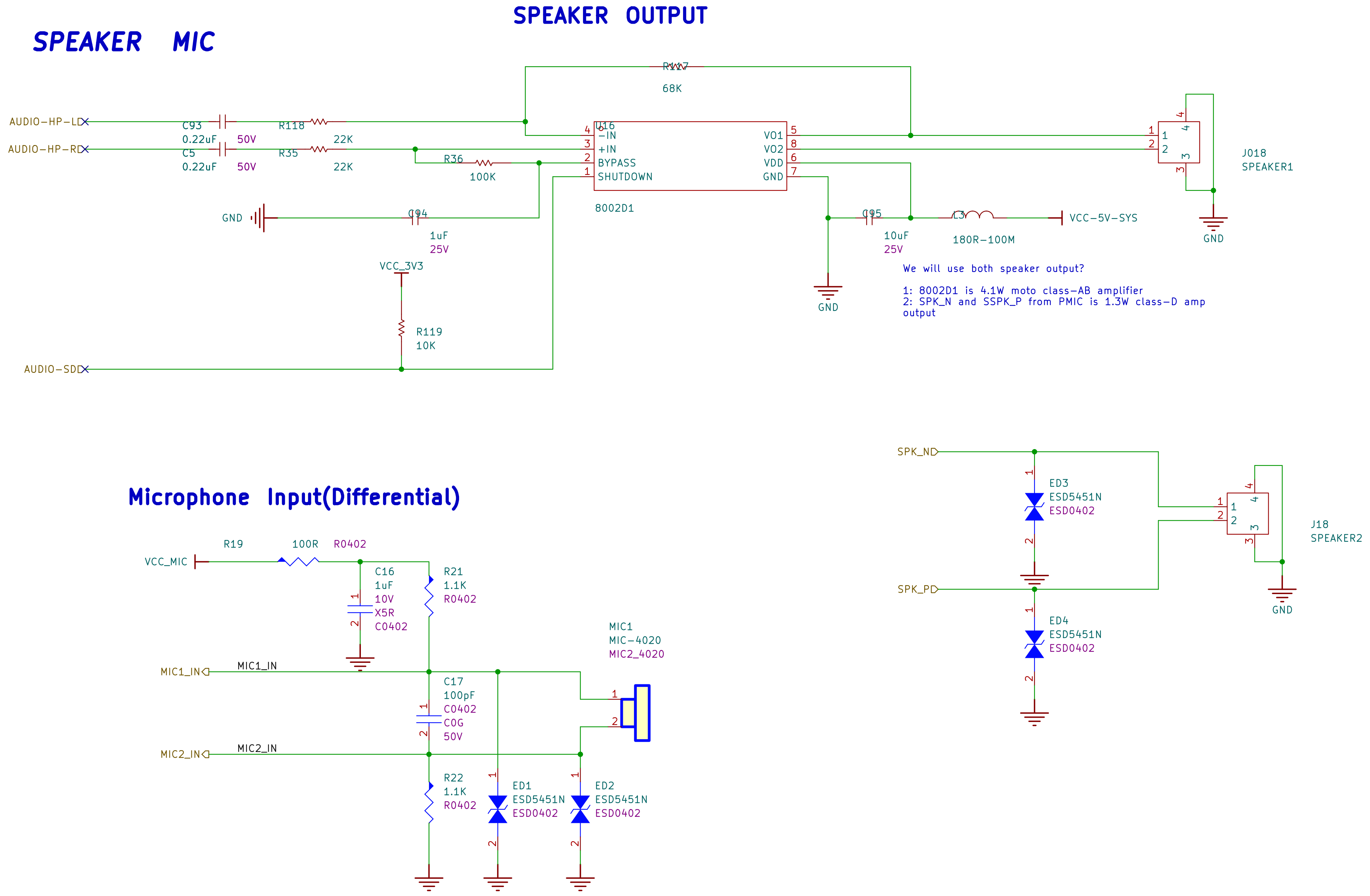
Title:

Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.3		Id: 25/32

LiDAR



Sheet: /LiDAR/ File: LiDAR .kicad_sch	
Title:	
Size: A2 Date:	**Rev:**
KiCad E.D.A. 10.0.3	Id: 26/32



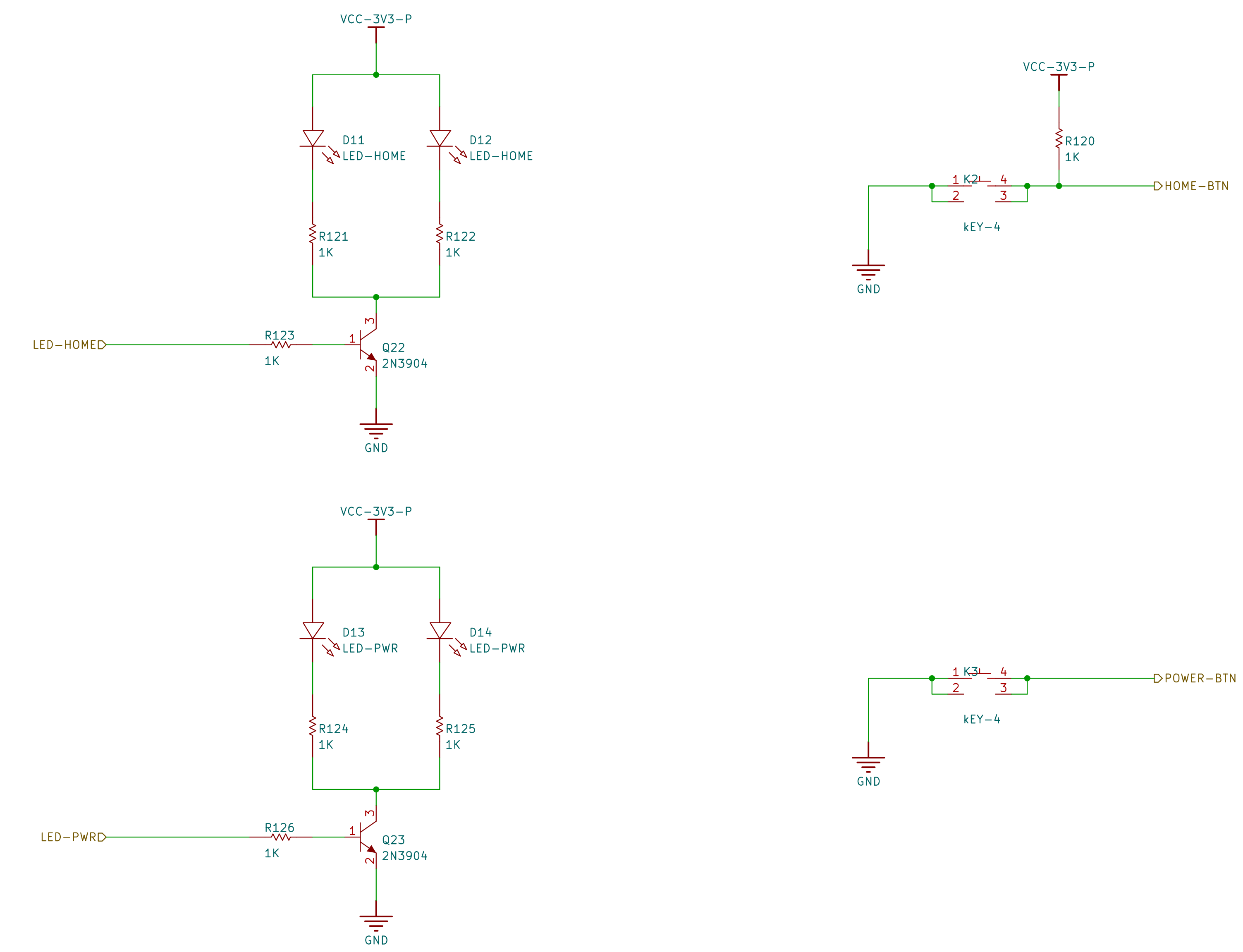
Sheet: /SPEAKER-MIC/
File: SPEAKER-MIC.kicad_sch

Title:

Size: A2 Date: Rev:

KiCad E.D.A. 10.0.3 Id: 27/32

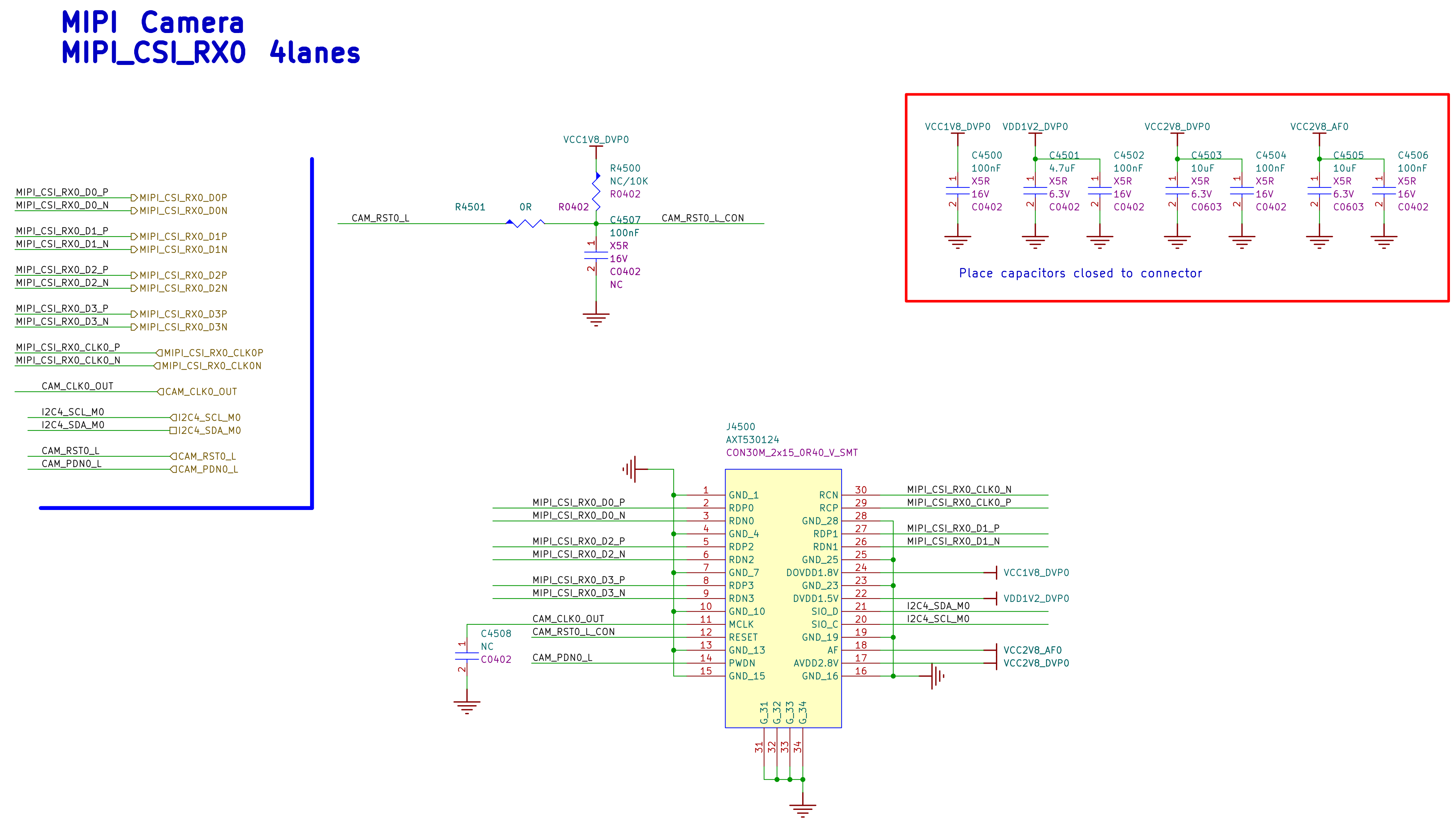
LEDs AND BUTTONs



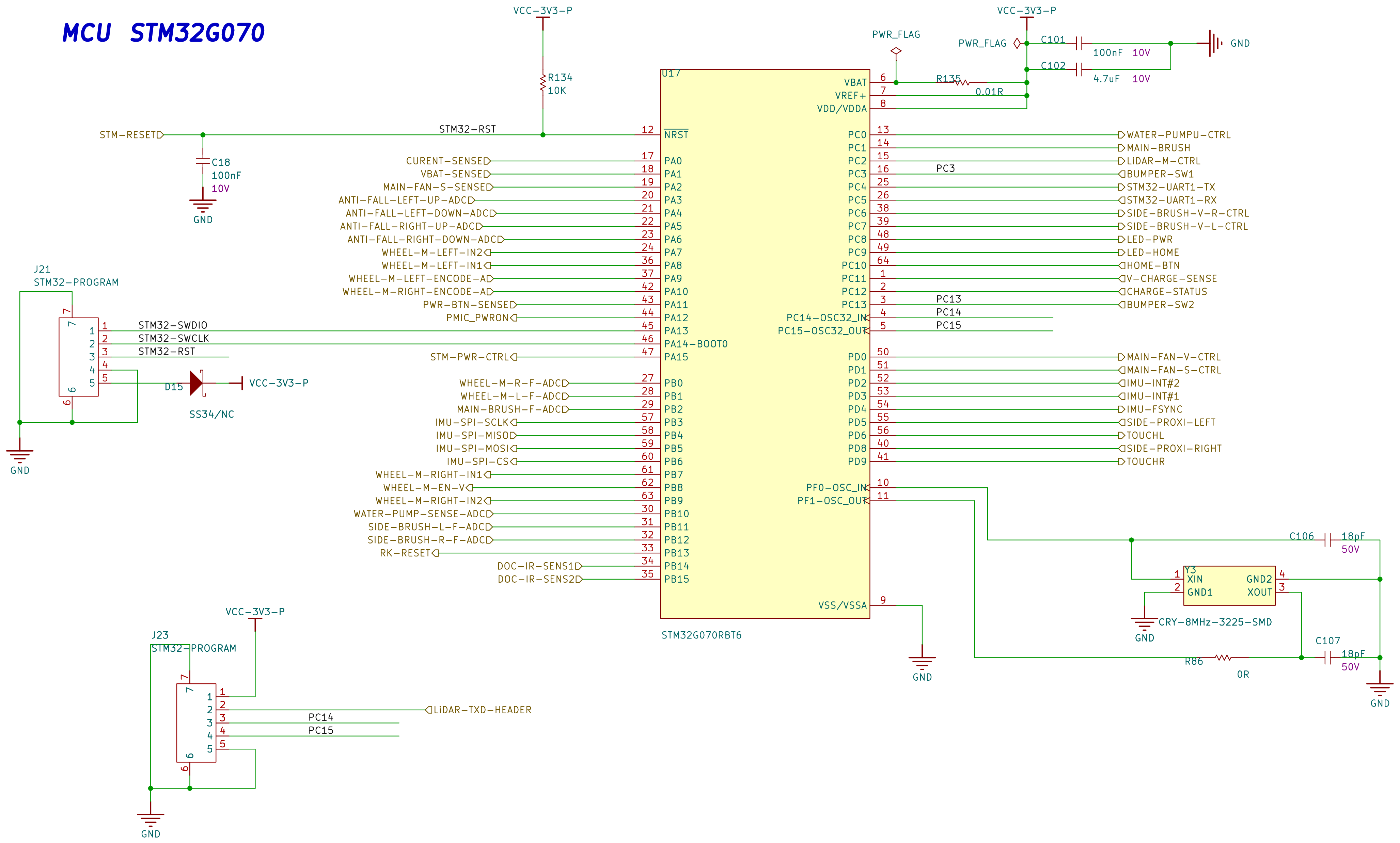
Sheet: /BUTTON-LED/
File: BUTTON-LEDs .kicad_sch

Title:

Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.3		Id: 28/32



Sheet: /CAMERA/
File: MIPI-CAMERA.kicad_sch
Title:
Size: A2 Date: Rev:
KiCad E.D.A. 10.0.3 Id: 29/32

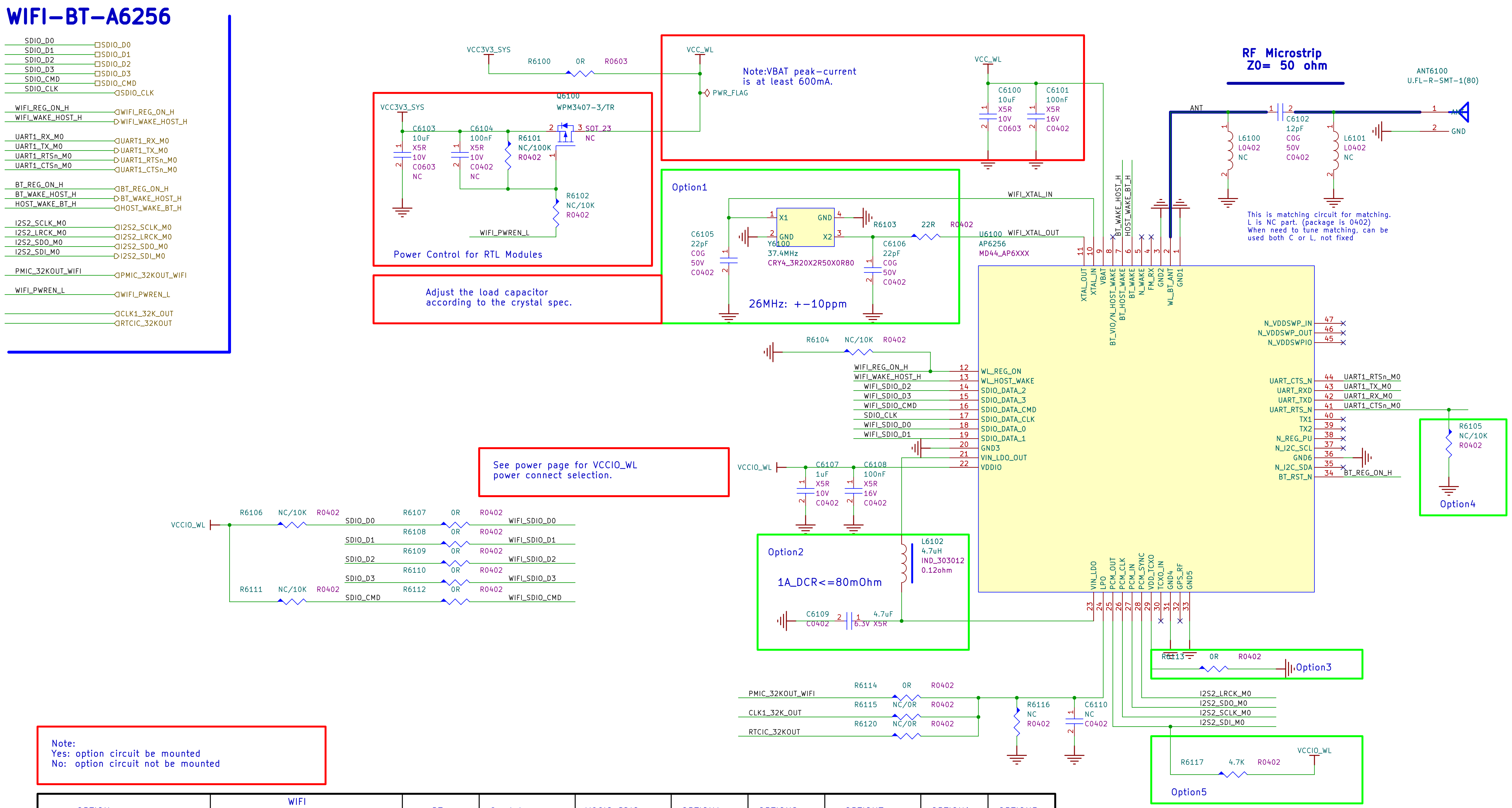


Sheet: /MCU-STM32/
File: STM32G070RBT6.kicad_sch

Title:

Size: A2 Date: Rev:

KiCad E.D.A. 10.0.3 Id: 30/32

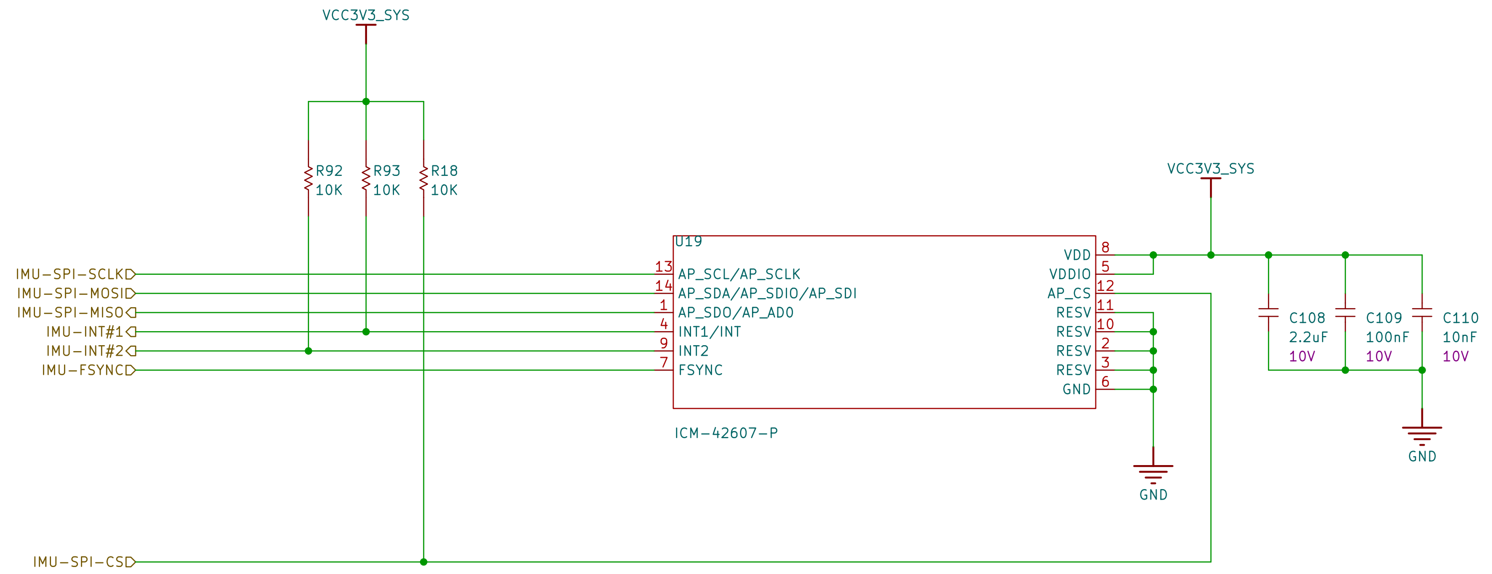


Note:
Yes: option circuit be mounted
No: option circuit not be mounted

OPTION	WIFI				BT	Crystals	VCCIO_SDIO	OPTION1	OPTION2	OPTION3	OPTION4	OPTION5
	a	b/g/n	ac	5GHz								
AW-CM256SM	Yes	Yes	Yes	Yes	4.2	37.4MHz	1.71-3.6V	Yes	Yes	Yes@SDIO2.0 No@SDIO3.0	No	No
AP6236/AP6212	No	Yes	No	No	4.2/4.0	26MHz	1.71-3.6V	Yes	Yes	No	No	No
AP6256/AP6255	Yes	Yes	Yes	Yes	5.0/4.2	37.4MHz	1.71-3.6V	Yes	Yes	Yes@SDIO2.0 No@SDIO3.0	No	No
RTL8189ETV Module F89FTSM12-W3	No	Yes	No	No	No	Module Integrated	1.8-3.3V	No	No	No	No	No
RTL8723DS Module 6223A-SRD	No	Yes	No	No	4.2	Module Integrated	1.62-3.6V	No	No	No	Yes	Yes
RTL8821CS Module 6221A-SRC	Yes	Yes	Yes	Yes	4.2	Module Integrated	1.7-3.45V	No	No	No	No	No

Sheet: /WIFI-BT/
File: WF-BT-AP6256.kicad_sch
Title:
Size: A2 Date: Rev:
KiCad E.D.A. 10.0.3 Id: 31/32

IMU-ICM-42607-P



Sheet: /IMU/
File: IMU-ICM-4267-P .kicad_sch

Title:

Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.3		Id: 32/32